

Ultrassonografia pulmonar simplificada: guia passo a passo

 pocus101.com/lung-ultrasound-made-easy-step-by-step-guide

Vi Dinh

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A ultrassonografia pulmonar (também conhecida como ultrassonografia torácica ou ultrassonografia do tórax) é uma aplicação de ultrassom à beira do leito (POCUS) que pode ser usada para diagnosticar quase qualquer patologia pulmonar em segundos, superando a radiografia de tórax!

Após ler este tutorial sobre ultrassom pulmonar, você aprenderá:

- Como realizar um protocolo de ultrassom pulmonar
- Posicionamento correto da sonda de ultrassom pulmonar
- Técnica de ultrassom pulmonar
- O que significa deslizamento pulmonar?
- O que significa “linhas B”?
- Como interpretar um ultrassom pulmonar normal
- Todos os sinais ultrassonográficos pulmonares com exemplos.
- Como diagnosticar todas as principais patologias pulmonares por ultrassom

[Imprima ou baixe a versão em PDF GRATUITA deste post sobre ultrassom pulmonar AQUI para ler quando quiser!](#)

Nota do editor: O protocolo de ultrassom pulmonar de 6 pontos que descrevemos neste post é o utilizado em nossa instituição e foi adaptado de [Lichtenstein, 2014](#). Mostraremos como identificamos todos os pontos para facilitar o acompanhamento. Sinta-se à vontade para adaptar ou revisar o protocolo de ultrassom pulmonar conforme achar melhor em sua instituição!

Índice

Indicações para ultrassonografia pulmonar

A ultrassonografia pulmonar pode ser usada na avaliação de:

- Falta de ar
- Pneumonia ou outras infecções recorrentes
- Edema pulmonar cardiogênico

- Síndrome da Angústia Respiratória Aguda (SARA)
- DPOC/Asma
- Pneumotórax

As principais limitações da ultrassonografia pulmonar são:

- Enfisema subcutâneo: as ondas de ultrassom não conseguem atravessar o ar na pele para avaliar os pulmões.
- Biotipo do paciente: pode ser difícil visualizar o pulmão em pacientes com obesidade grave.

Preparação para o exame de ultrassom pulmonar

Preparação do Paciente

- **Posição do paciente:** o paciente pode estar em decúbito dorsal, em decúbito lateral ou sentado. Para esta publicação, o paciente estará em decúbito dorsal.
- Se você estiver examinando o pulmão direito, peça ao paciente que levante o braço direito e coloque a mão atrás da cabeça, e vice-versa para o pulmão esquerdo.



Decúbito dorsal com o braço levantado



Em pé

Preparação do aparelho de ultrassom

Uma dúvida frequente é: "Qual transdutor de ultrassom é usado para ultrassom pulmonar?" Em geral, um transdutor curvilíneo ou um transdutor phased array são suficientes. No entanto, se você realmente deseja visualizar a pleura superficial, pode ser necessário usar um transdutor linear.

- **Transdutor** : [Sonda ultrassônica curvilínea](#) ou [sonda phased array](#) .
O [transdutor linear](#) pode ser usado para imagens superficiais, permitindo examinar a pleura próxima à parede torácica.
- **Pré-configuração** : Pulmão ou Abdômen
- **Posicionamento do aparelho** : Posicione o aparelho no lado direito do paciente, de forma que você possa realizar o exame com a mão direita e manipular os botões do ultrassom com a mão esquerda.



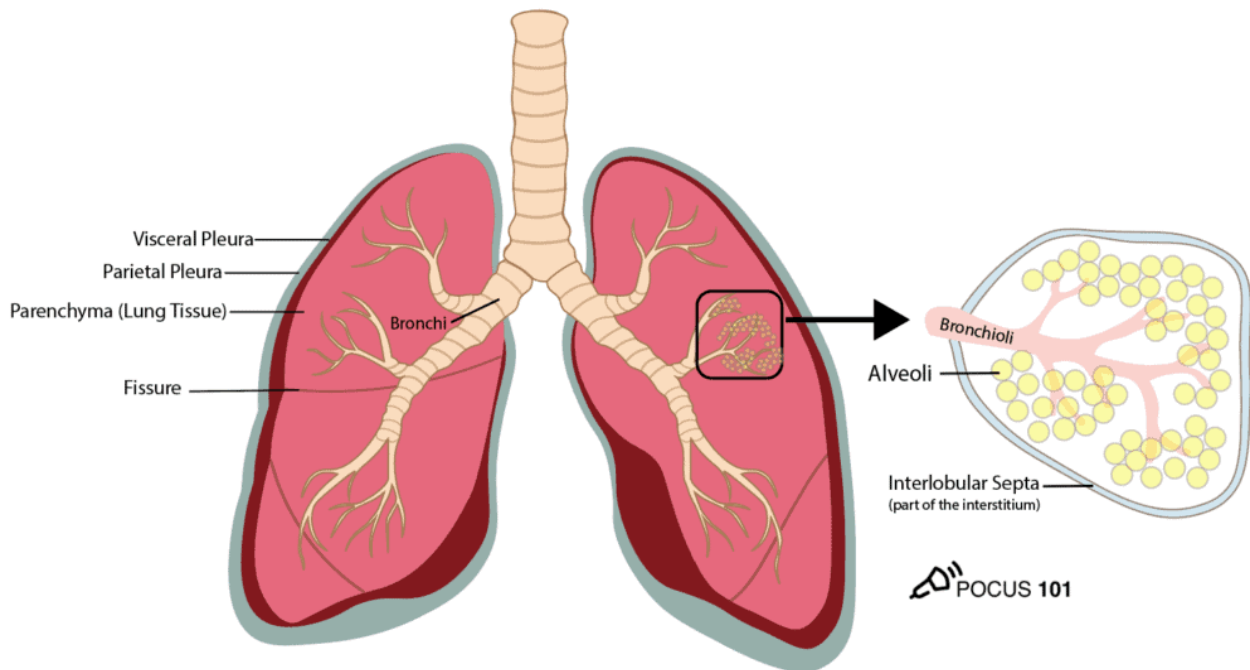
Anatomia pulmonar por ultrassom

Usaremos ultrassom pulmonar para procurar patologias que afetem a pleura, os alvéolos e o interstício.

A **pleura parietal** se interconecta com a **pleura visceral** , criando um movimento de deslizamento durante a respiração. Mais adiante, discutiremos como esse movimento de

"deslizamento pulmonar" é um achado muito importante durante um ultrassom, pois pode descartar processos patológicos como o pneumotórax.

Os alvéolos existem em lóbulos subdivididos por **septos interlobulares**. Os septos se ancoram na pleura visceral para estabilizar os lóbulos. Quando essas áreas se enchem de líquido devido à consolidação ou edema pulmonar, observamos diversos artefatos na ultrassonografia.



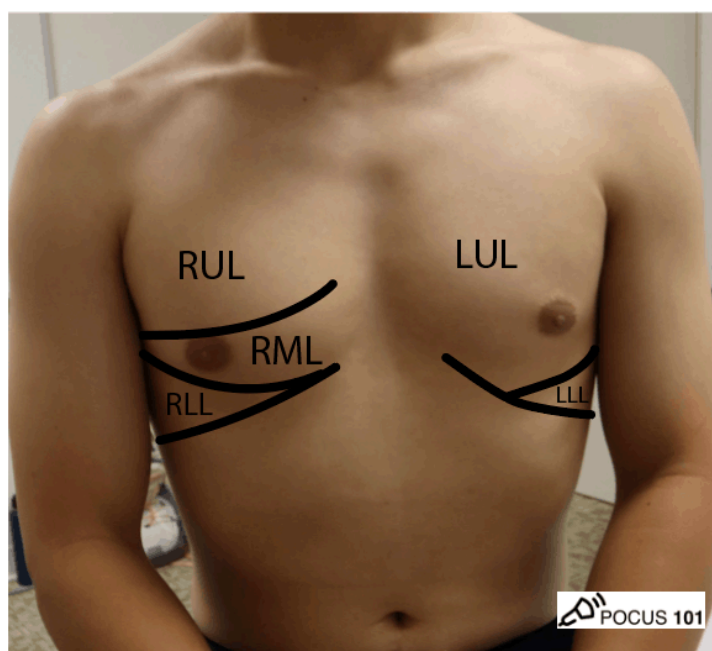
É importante lembrar que as ondas de ultrassom são completamente dispersas e atenuadas (absorvidas) pelo ar que preenche os alvéolos saudáveis. Portanto, se o paciente tiver pulmões normais, você não deverá conseguir visualizar a textura do parênquima (tecido pulmonar) durante o exame.

No entanto, em certas patologias, como o edema pulmonar intersticial, o acúmulo de líquido nos septos interlobulares resulta em artefatos no ultrassom pulmonar conhecidos como "linhas B". E, à medida que a doença pulmonar progride, pode-se começar a observar até mesmo consolidação pulmonar completa no ultrassom. Discutiremos isso mais detalhadamente na seção de patologia deste artigo.

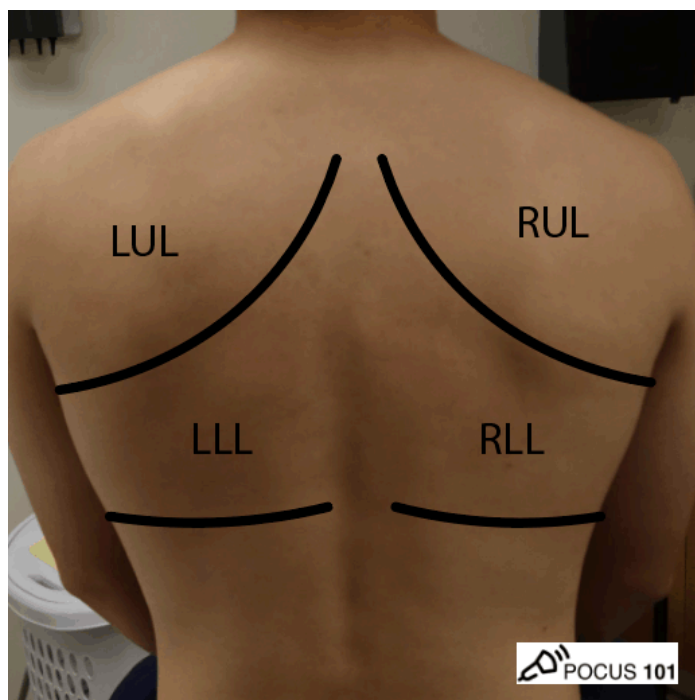
Anatomia de superfície do pulmão e dos lobos

Aqui está uma breve revisão da anatomia da superfície pulmonar. Aqui podemos ver a anatomia da superfície dos lobos superiores direito e esquerdo, do lobo médio direito e do lobo inferior esquerdo. Na próxima seção, explicaremos o exame de ultrassom pulmonar de 6 pontos que utilizamos.

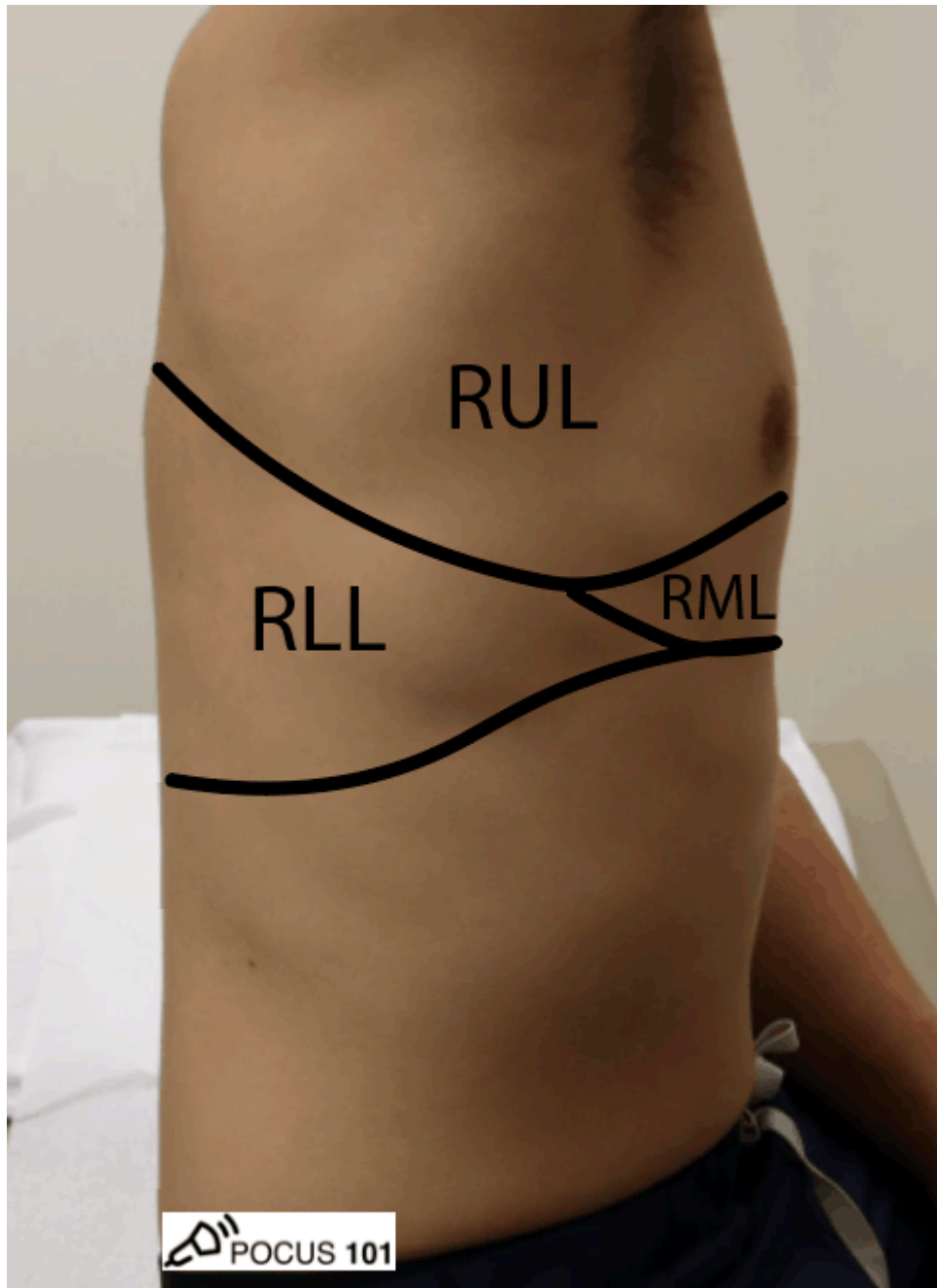
Abreviações : RUL: Lobo Superior Direito, RML: Lobo Médio Direito, RLL: Lobo Inferior Direito, LUL: Lobo Superior Esquerdo, LLL: Lobo Inferior Esquerdo



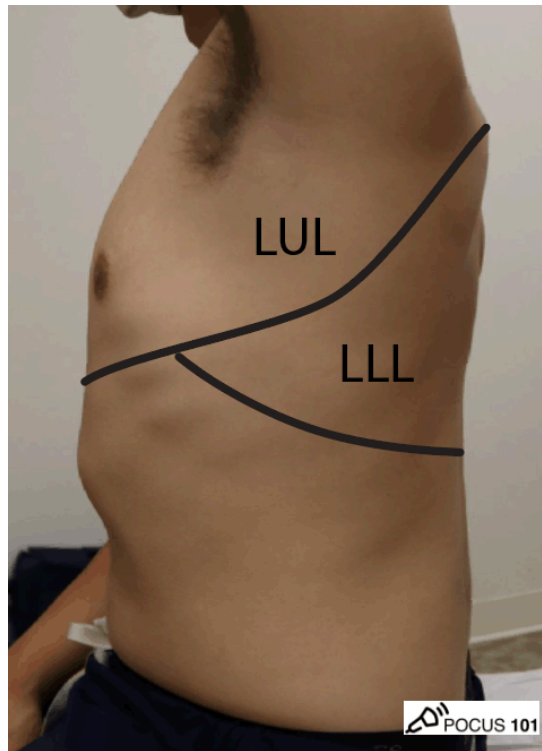
Anatomia da Superfície Pulmonar – Anterior



Anatomia da Superfície Pulmonar – Posterior



Anatomia da superfície pulmonar – Vista lateral direita



Anatomia da superfície pulmonar – Vista lateral esquerda

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Protocolo passo a passo para ultrassom pulmonar

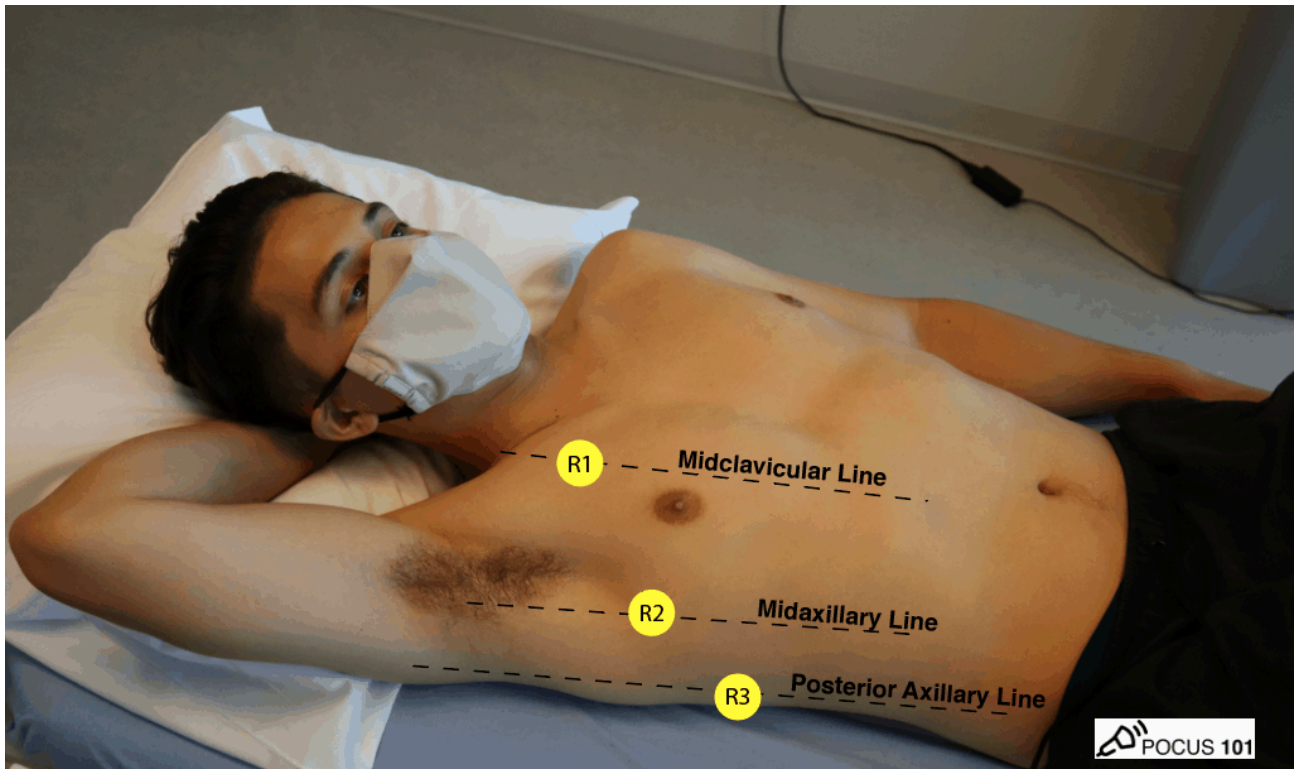
Definindo o exame de ultrassom pulmonar de 6 pontos

Em geral, recomendamos um exame de 6 pontos para ultrassonografia pulmonar geral ([Lichtenstein, 2014](#)). No entanto, em situações de avaliação de COVID-19 ou outras pneumonias virais, o pulmão pode ser afetado de forma multilobar, sendo recomendável considerar um exame mais abrangente de 12 pontos, que você pode [ler AQUI](#) .

Nesta seção, você aprenderá como posicionar corretamente as sondas e identificar as estruturas básicas. Em nossa instituição, identificamos os 6 pontos pulmonares para facilitar a referência e a identificação nos aparelhos de ultrassom. Referimo-nos a esses pontos como R1, R2 e R3 para o pulmão direito e L1, L2 e L3 para o pulmão esquerdo, respectivamente.

Posição da sonda	Campo pulmonar
R1 e L1	Anterior Superior
R2 e L2	Lateral
R3 e L3 (PLAPS)	Posterior Inferior

Rotulagem por ultrassom pulmonar



Pontos de ultrassom pulmonar para um protocolo de 6 pontos

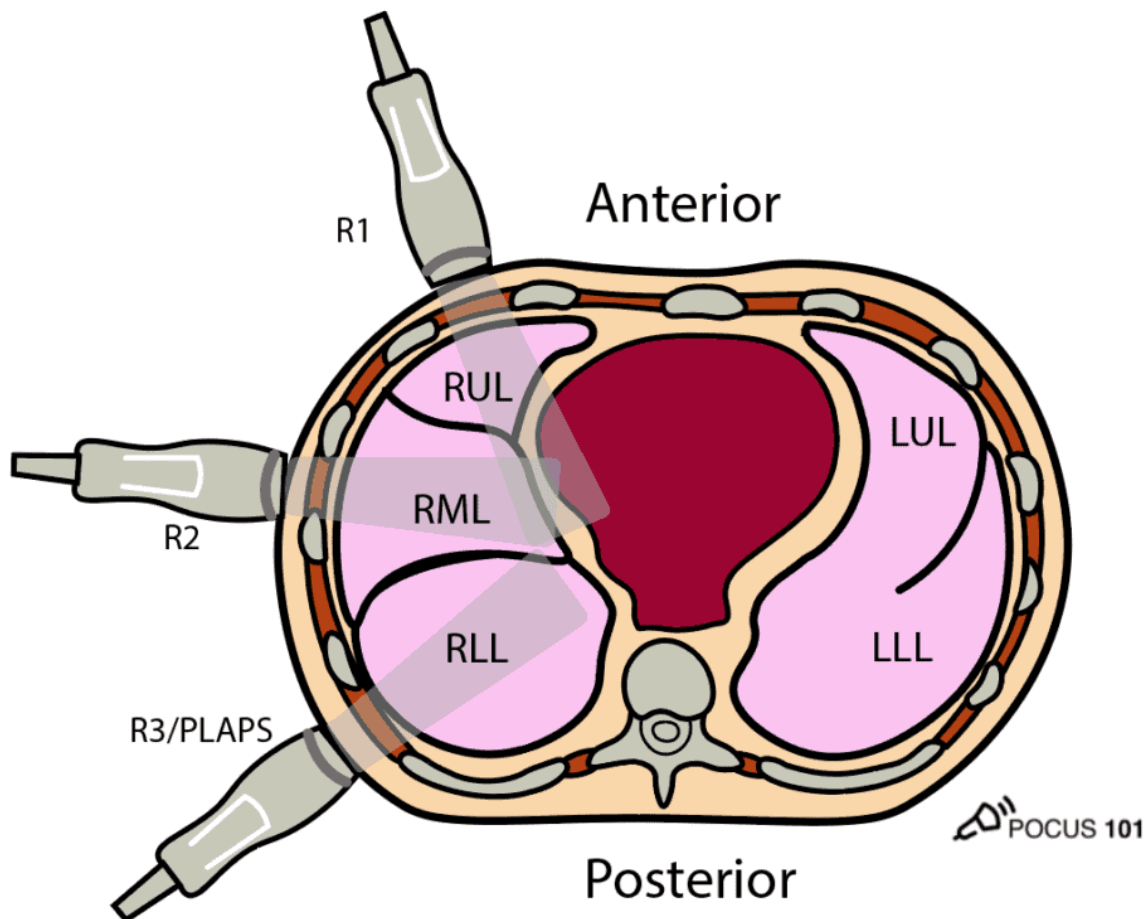
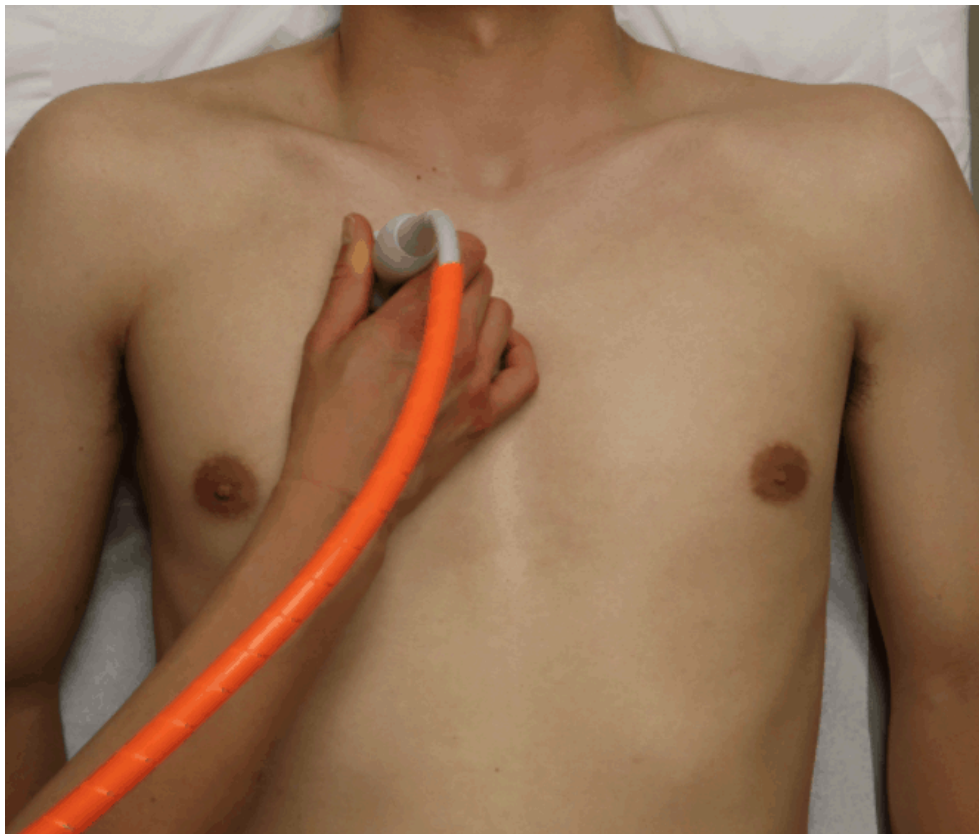


Ilustração ultrassonográfica pulmonar em vista transversal.

Passo 1 : Ultrassonografia pulmonar da região anterior do tórax (D1 / E1)

O ponto 1 (R1 e L1) avalia a região anterior do tórax. Este ponto será mais relevante na avaliação de **pneumotórax e/ou edema intersticial**.

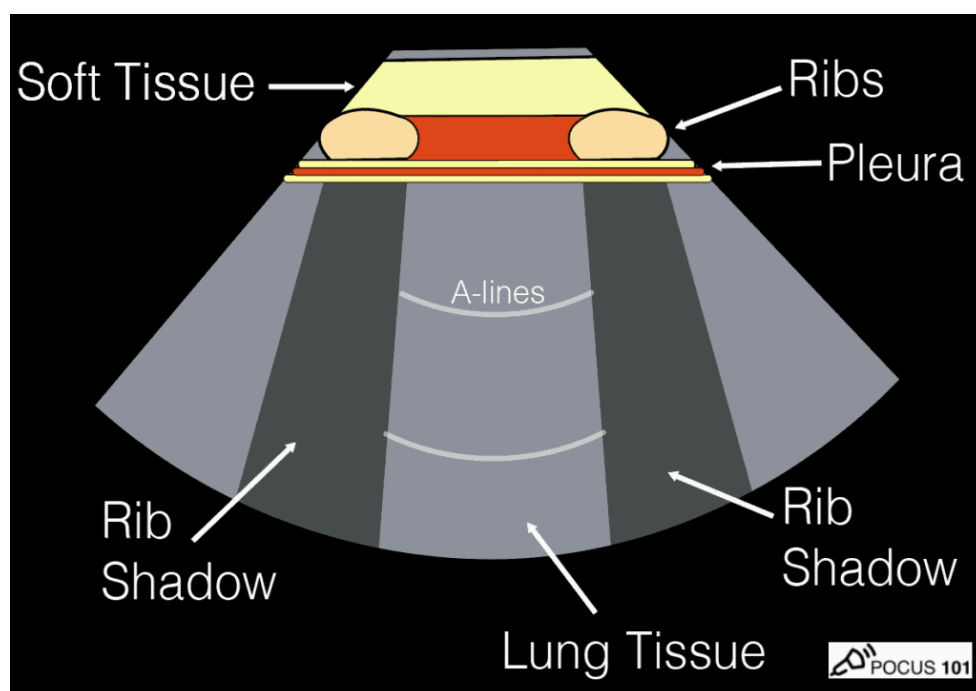
- Aponte o indicador em direção à cabeça do paciente.
- Posicione a sonda na **linha hemiclavicular, no segundo espaço intercostal do pulmão direito (R1) e esquerdo (L1), respectivamente**.
- Posicione a sonda no espaço entre duas costelas.

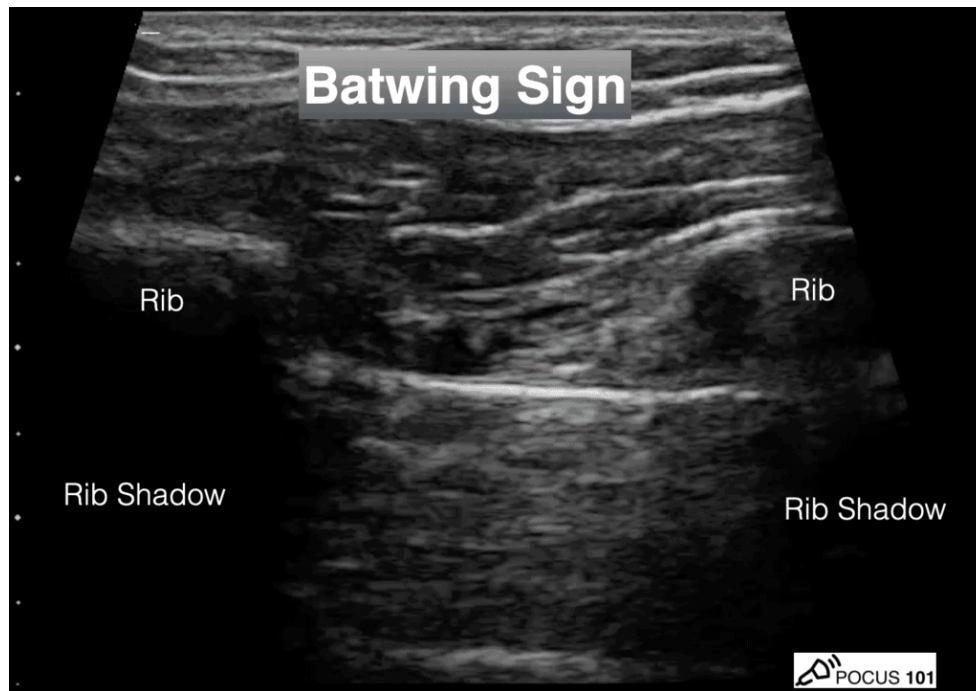


R1 Ultrasound Probe Position

Identify Two Rib Shadows (Batwing Sign)

The first lung ultrasound finding to confirm you are in the correct position is to look for the two rib shadows or the “**Batwing Sign**.” This ensures that your probe is in between two ribs.





Batwing Sign on Ultrasound

Identify Lung Sliding

- The next finding you will want to look for is lung sliding.
- Lung sliding is a normal finding where the visceral and parietal pleura slide back and forth on one another as the patient breathes. Some say this looks like tiny “ants marching on a line.”
- This is a simple finding but extremely useful since lung sliding definitely means that the visceral and parietal pleura are next to each other, effectively ruling out pathology such as pneumothorax.

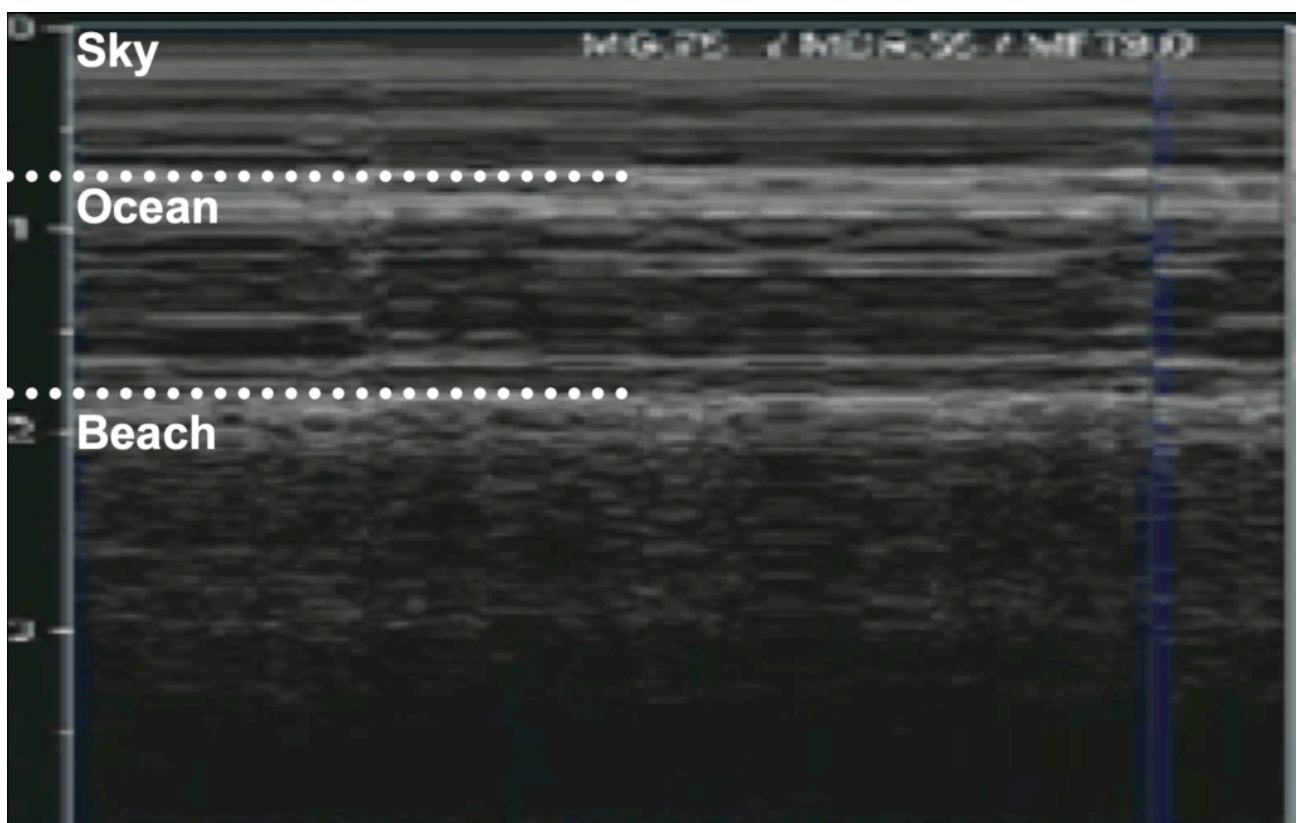
Editor's Note: Lung sliding can be seen with the phased array and curvilinear ultrasound probes. However it is often times most easily seen with the linear probe given how shallow it is.



Lung sliding with linear probe

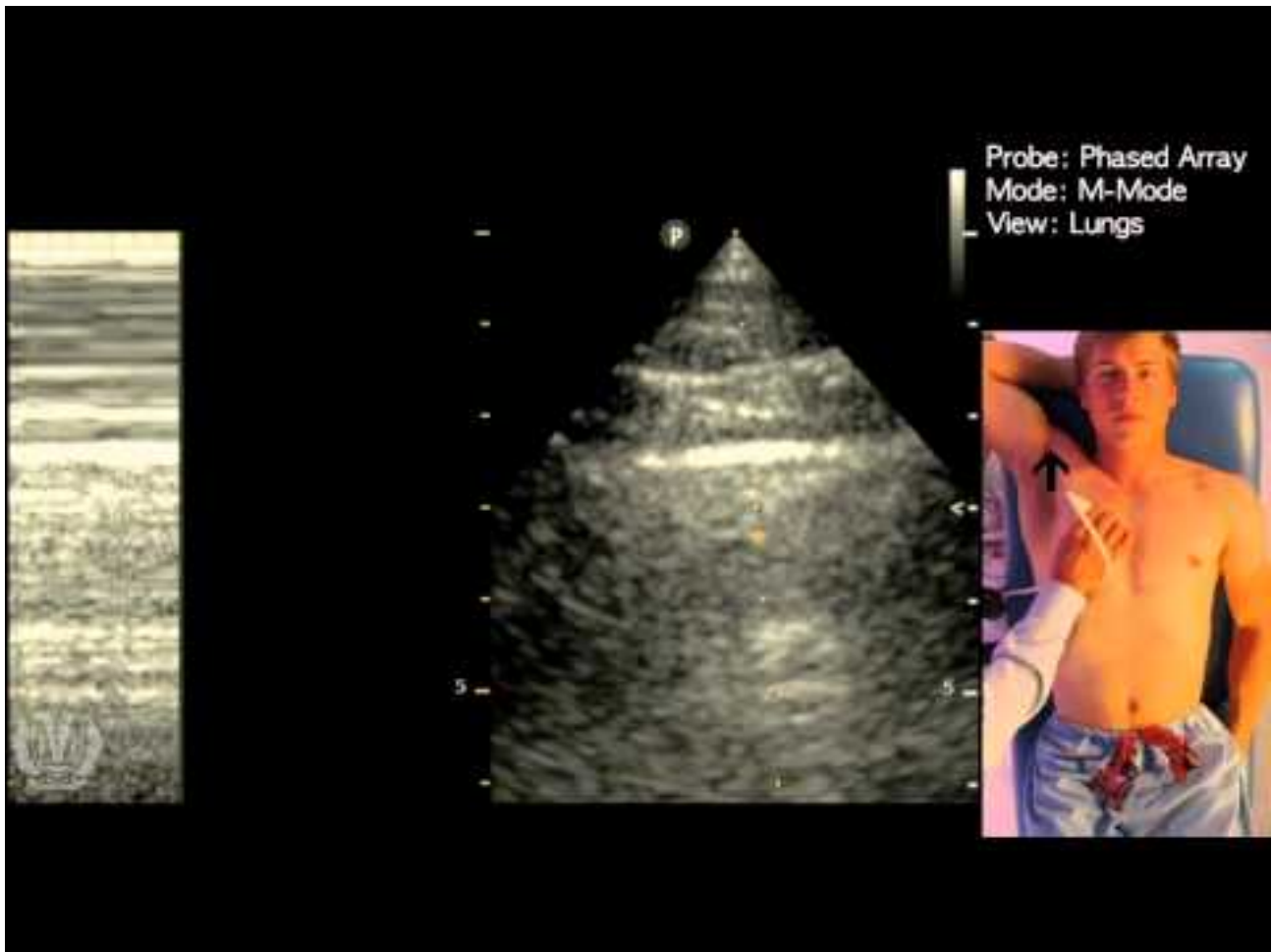
Lung Sliding using M-Mode (Optional)

- If lung sliding is not readily apparent, it can be further evaluated using [M-Mode](#). The goal of M-Mode is to see if the patient has a **normal seashore sign**.
- **Seashore Sign:** Sky = Skin/Subcutaneous Tissue, Ocean = Muscle, Beach = Lung sliding motion (sandy appearance)



Seashore Sign on Lung Ultrasound

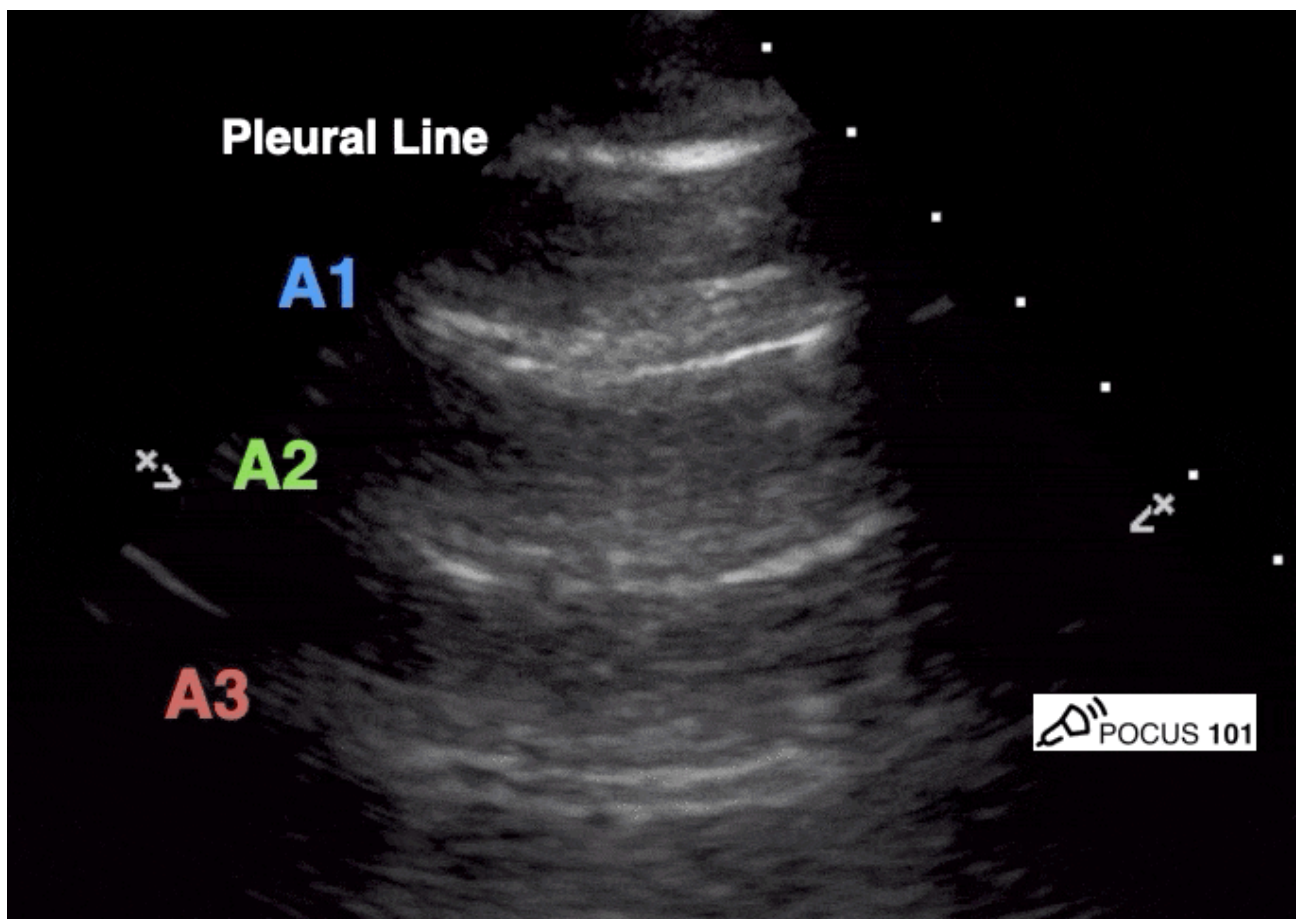
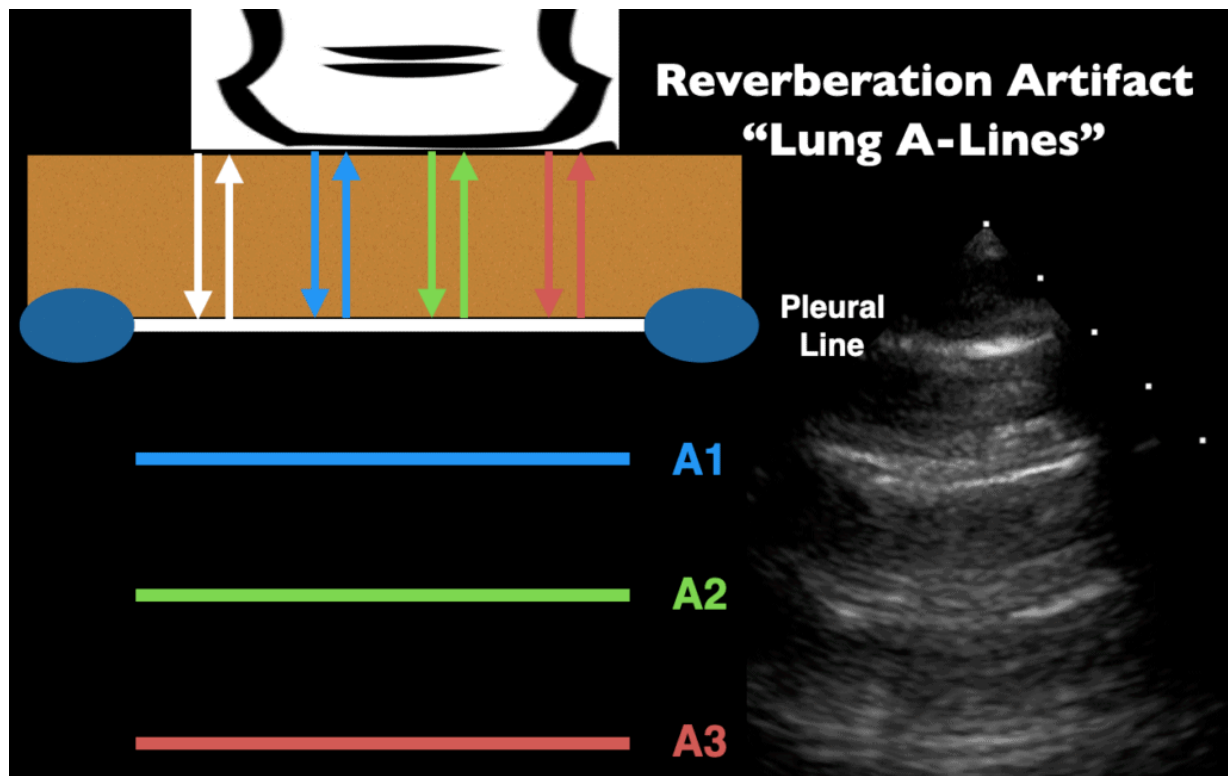
Here is a quick video on how to use M-mode for lung ultrasound:



[Watch on YouTube](#)

Identify A-lines

When you angle your probe **perpendicular to the pleura**, ultrasound waves reflect off the air from the pleura and the probe surface, producing an image of what looks like pleural lines that lie equidistant to the true pleural line. These reverberation artifacts can continue reflecting back and cause several **A-lines** to appear before dissipating. (A1 = first A-line seen, A2 = second A-line seen, A3 = third A-line seen)



Clinically, A-lines can indicate healthy aeration of the alveoli. However, A-lines can also be seen in cases of pneumothorax since there will be a reflection of air at the parietal pleura. The difference is that you will not see lung sliding in pneumothorax (will discuss this much more in the pathology section below).

Step 2: Lung Ultrasound of LATERAL Chest (R2/L2)

Point 2 (R2 and L2) examines the lateral chest.

- Place your probe at **the right (R2) or left (L2) Midaxillary line around the 6-7th intercostal space**. This should be just lateral to the nipple line in males.
- Anchor your probe between two ribs, and just like in point 1, look for the **batwing sign, lung sliding, and A-lines**.



R2 Ultrasound Probe Position

Step 3: Lung Ultrasound of POSTERIOR Chest (R3/L3-PLAPS)

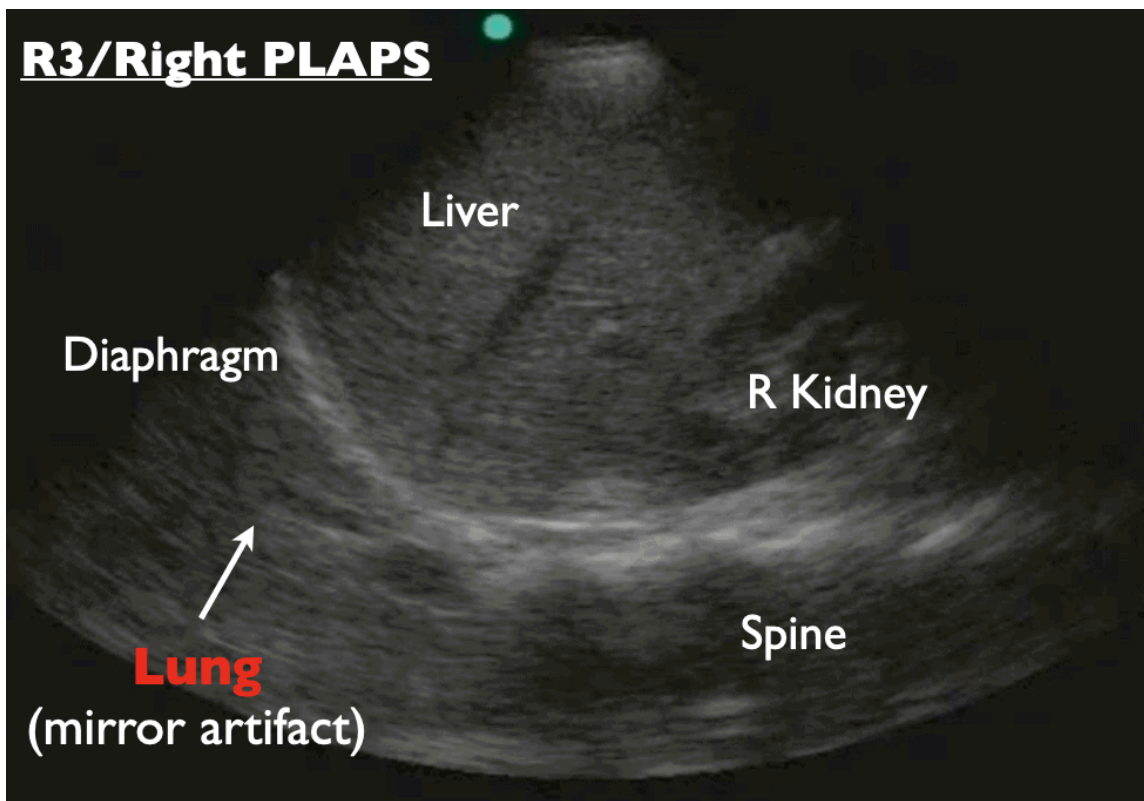
Point 3 (or R3 and L3) assesses the posterior chest. This point is commonly known as the **PLAPS** point on lung ultrasound (“posterior and/or lateral alveolar and/or pleural syndrome”). The PLAPS will be most relevant for assessing the presence of **pleural effusions and consolidations**.

- Point your indicator towards the patient’s head.
- Slide the probe under the patient at the PLAPS point at the **intersection of the posterior axillary line and a rib space between the 10th and 12th ribs**.

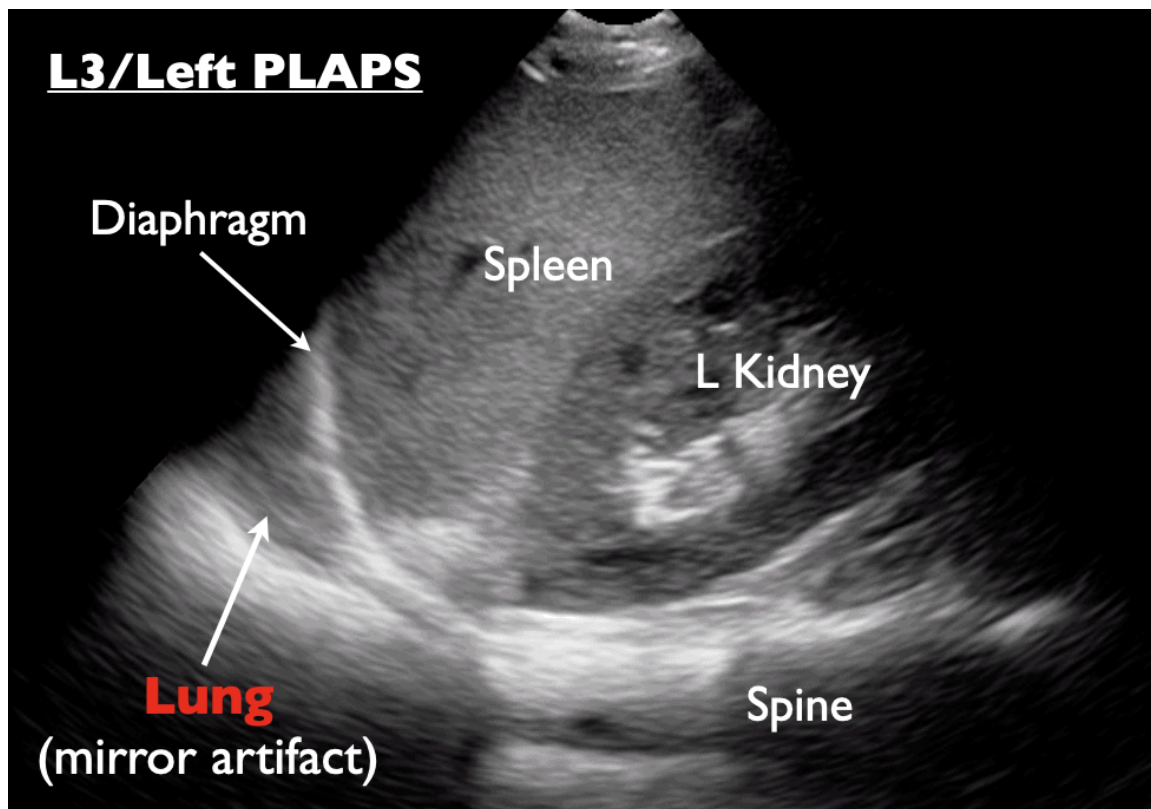


R3 – PLAPS Point

- Identify the **liver** (right side) or **spleen** (left side), **kidney**, and **diaphragm**.
These organs will come in and out of view when the lung inflates and deflates.
- You should also see the **spine**, which in healthy lungs, extends up only until the border of the diaphragm. If it extends past the diaphragm called the “spine sign”, you will need to look for a pleural effusion or consolidation which we explain how to do in the pathology section.



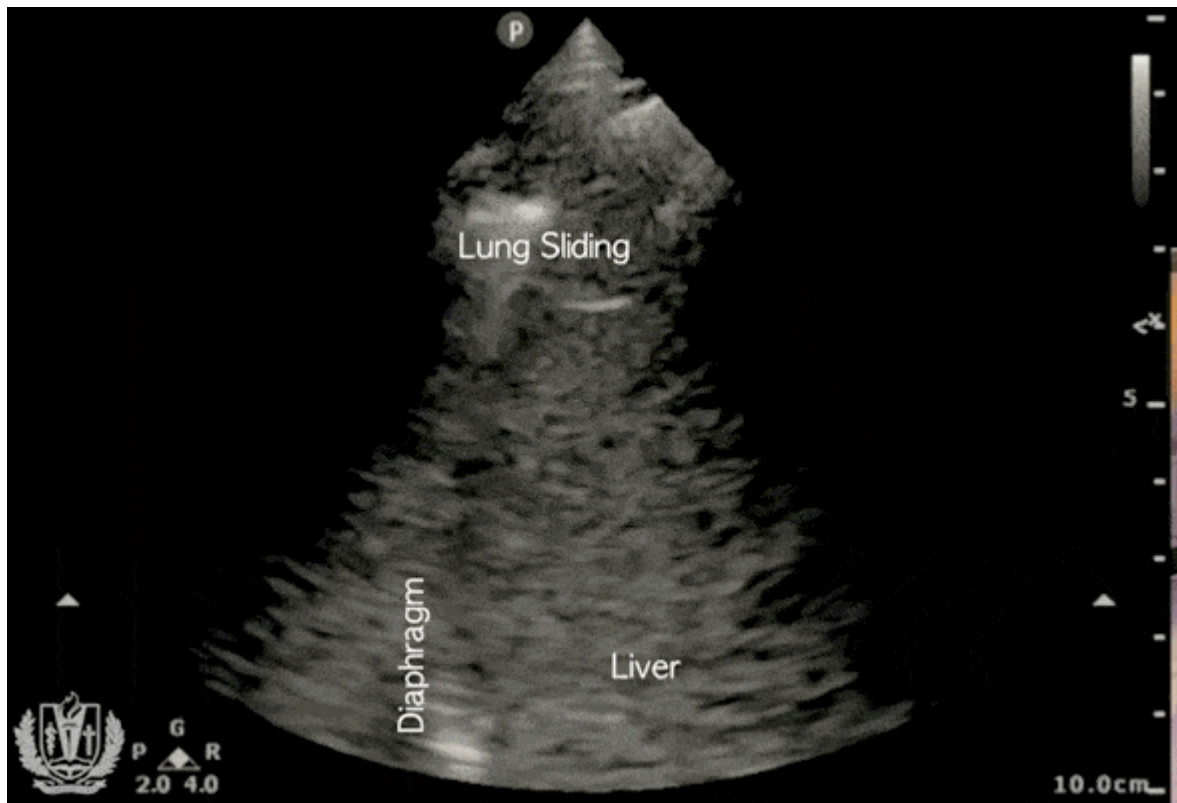
R3 (Right PLAPS point)



L3 (Left PLAPS point)

Curtain Sign

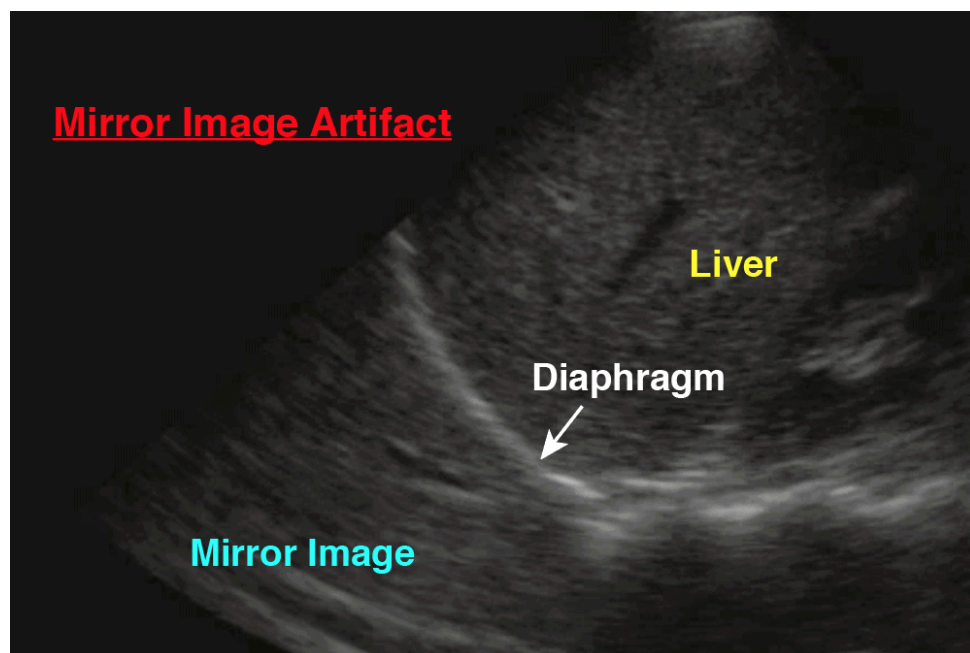
The curtain sign is seen in healthy and aerated lungs at point 3 (PLAPS position). An aerated lung is like a “curtain” because as it fills with air, it looks like a curtain sweeping down and over the other organs, momentarily obscuring them from view. The diaphragm, liver, or spleen reappear during exhale.



Inhalation causes curtain sign at the costodiaphragmatic angle

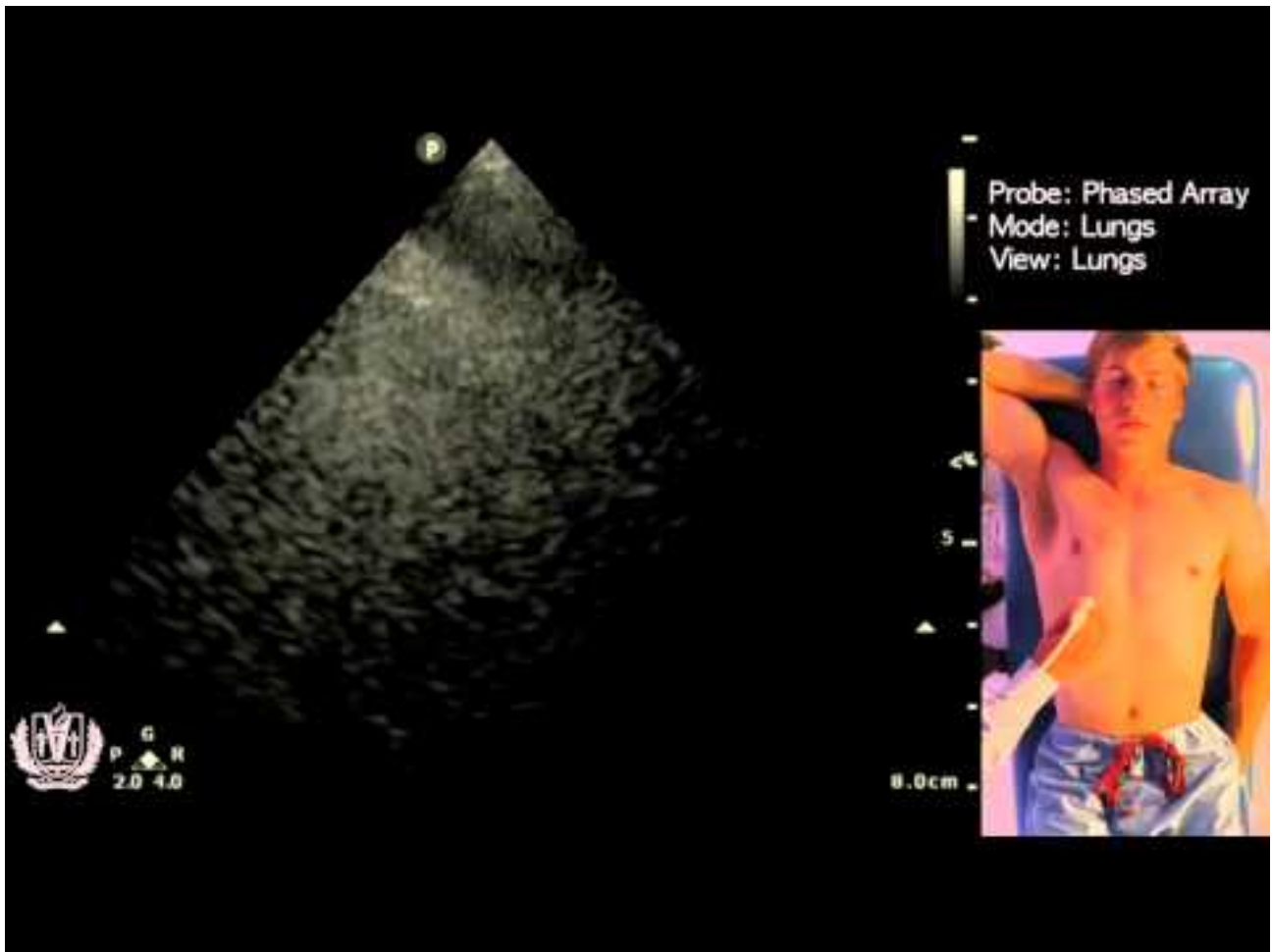
Mirror Image Artifact

[Mirror image artifacts](#) are a normal finding that occurs when reflection from a structure creates a false image behind a reflective object, such as the diaphragm, as seen in the image below. This is a normal finding in the PLAPS position.



Mirror Image Artifact

Lung Ultrasound Video Summary



[Watch on YouTube](#)

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Lung Ultrasound Signs and Findings

Before we look at specific lung diseases with ultrasound, you will first need to understand the pathological signs and findings of lung ultrasound.

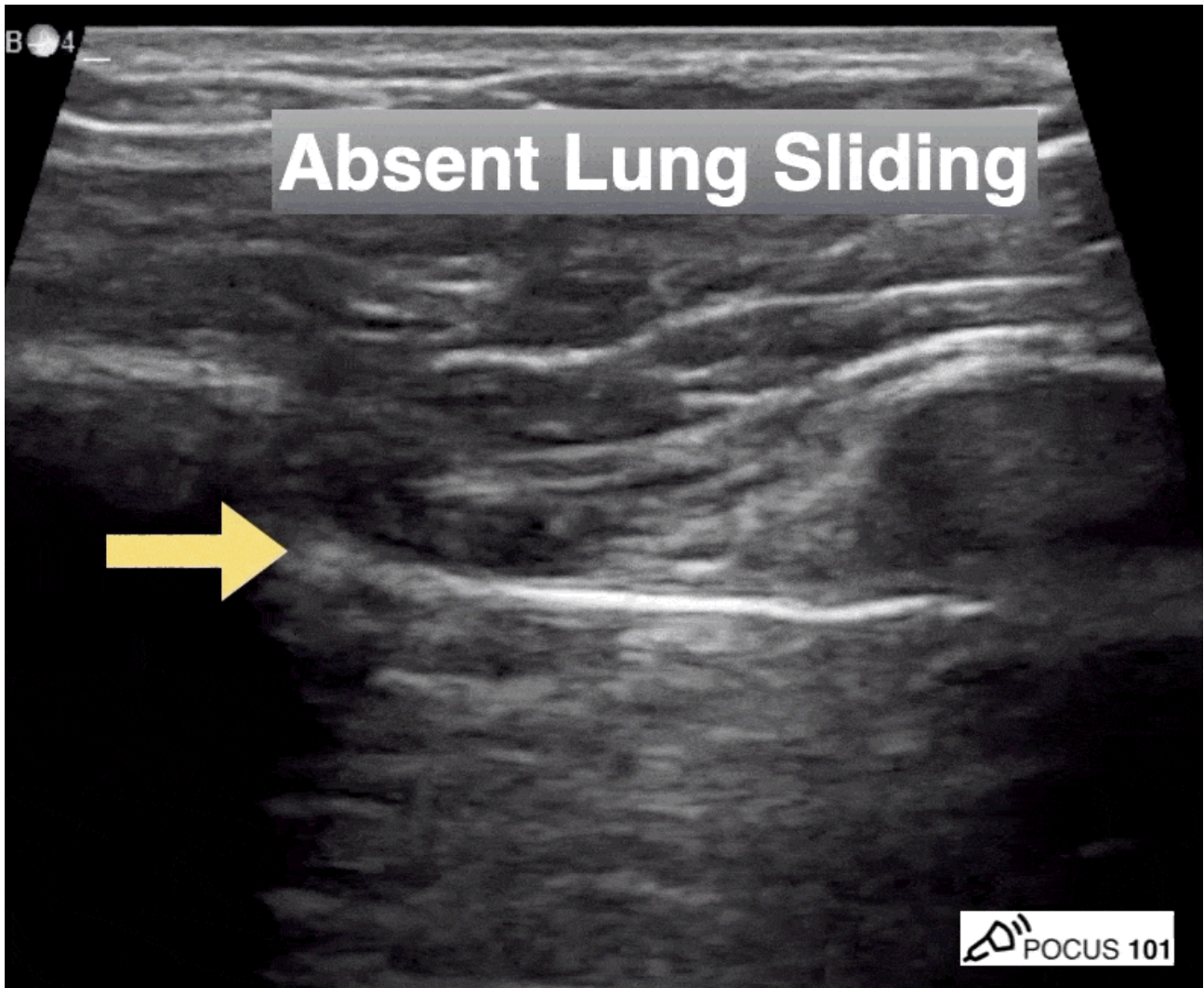
Lung ultrasound pathology profiles such as pneumothorax, pneumonia, cardiogenic pulmonary edema, etc will have a different combination and distribution of these pathological lung ultrasound findings/signs. Once you understand these basic lung ultrasound findings you will be able to interpret just about any lung ultrasound images.

Absent Lung Sliding

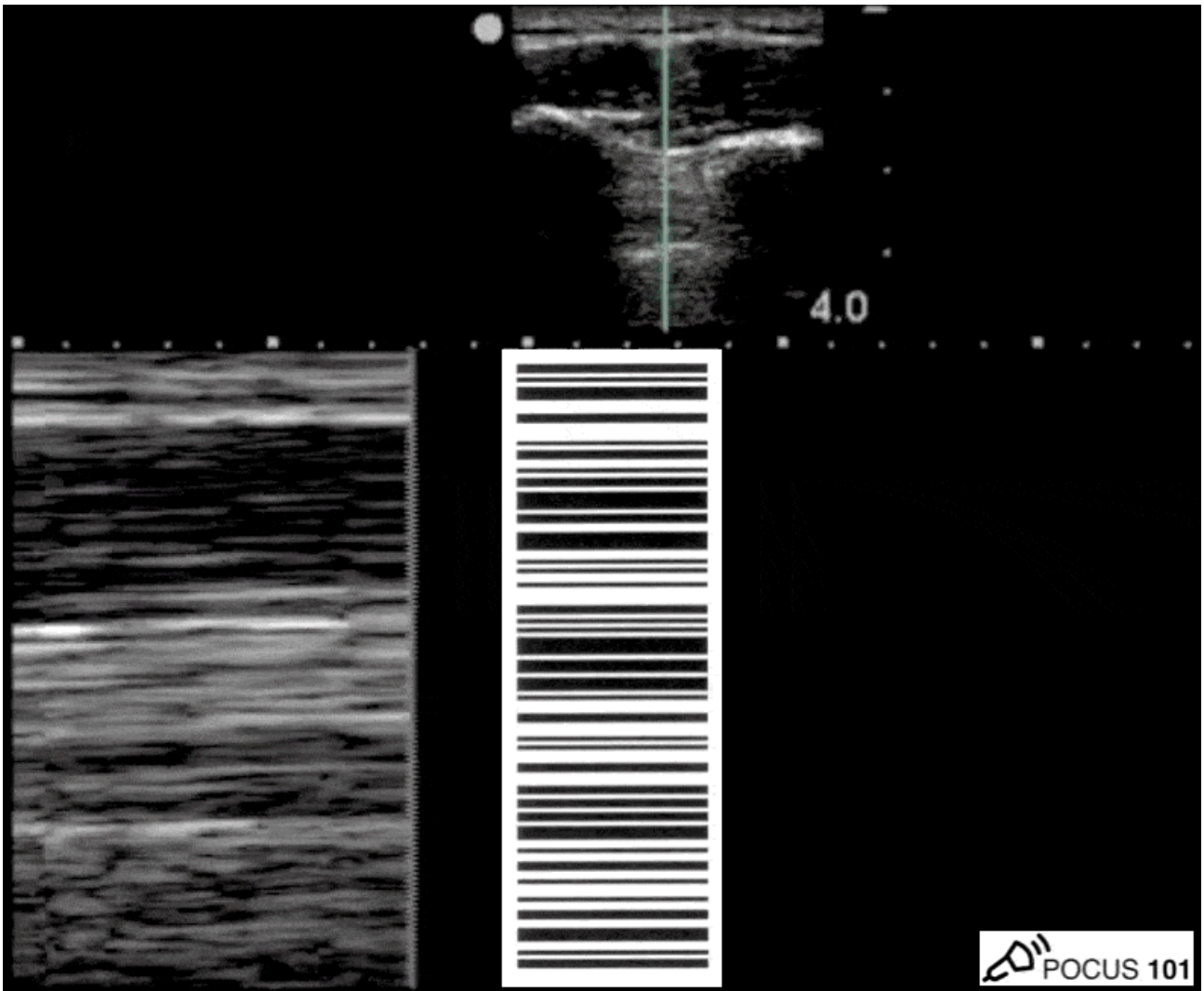
Lung sliding may be reduced or absent in a variety of circumstances. There is no lung sliding present when the parietal and visceral pleura become separated by air

(pneumothorax) or fluid (pleural effusion). Furthermore, severe COPD or anything that hyperinflates the lungs can markedly reduce pleural sliding.

Absence of lung sliding can be seen using B-mode or using M-mode (**stratosphere sign** and **barcode sign**).



Absence of Lung Sliding – B Mode



Absence of Lung Sliding – M Mode

The most clinically relevant fact about lung sliding is that **if lung sliding is present, you can rule out pneumothorax with 100% sensitivity**. On the flip side, the absence of lung sliding can have many causes with pneumothorax being one of them.

Thus, if your patient does not have lung sliding, you must use your patient's clinical features to consider other causes such as severe consolidation, chemical pleurodesis, acute infectious or inflammatory states, fibrotic lung diseases, acute respiratory distress syndrome, or mainstem intubation.

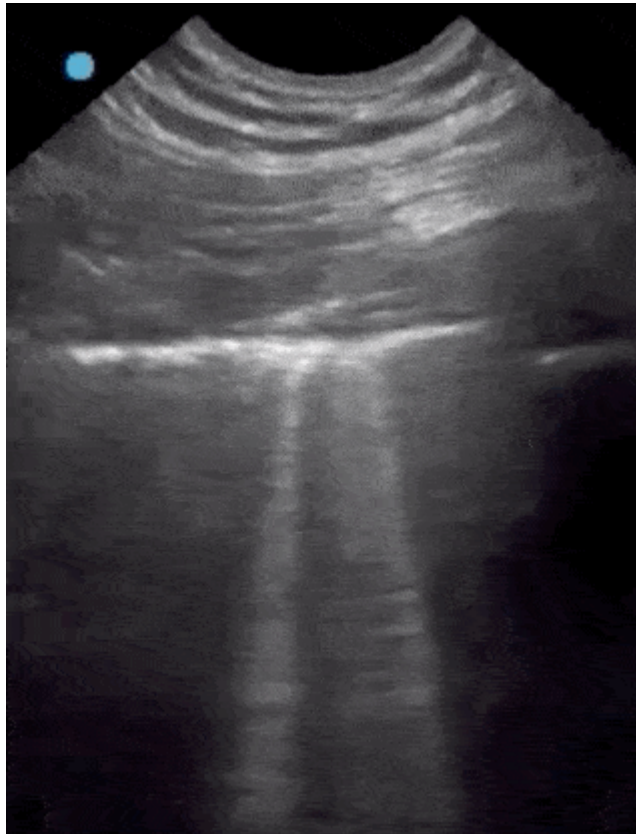
Interstitial Edema/Syndrome and B-lines

The lung interstitium, or the network of tissue that supports the alveoli and blood vessels, can thicken or fill with fluid.

B-lines form when interlobular septa and lung tissue thicken or fill with fluid. Thus, many clinicians have equated B-lines with “wet lung.” Remember, **A-lines are horizontal while B-lines are vertical**.

In order to be sure you are seeing B-lines, make sure they have these following qualities (Lichtenstein, et al):

- Appear ray-like, hyperechoic, and vertical.
- Emanate from the pleural line.
- Move with lung sliding.
- Extend to the periphery of the far-field.
- Can be associated with a thickened pleural membrane.

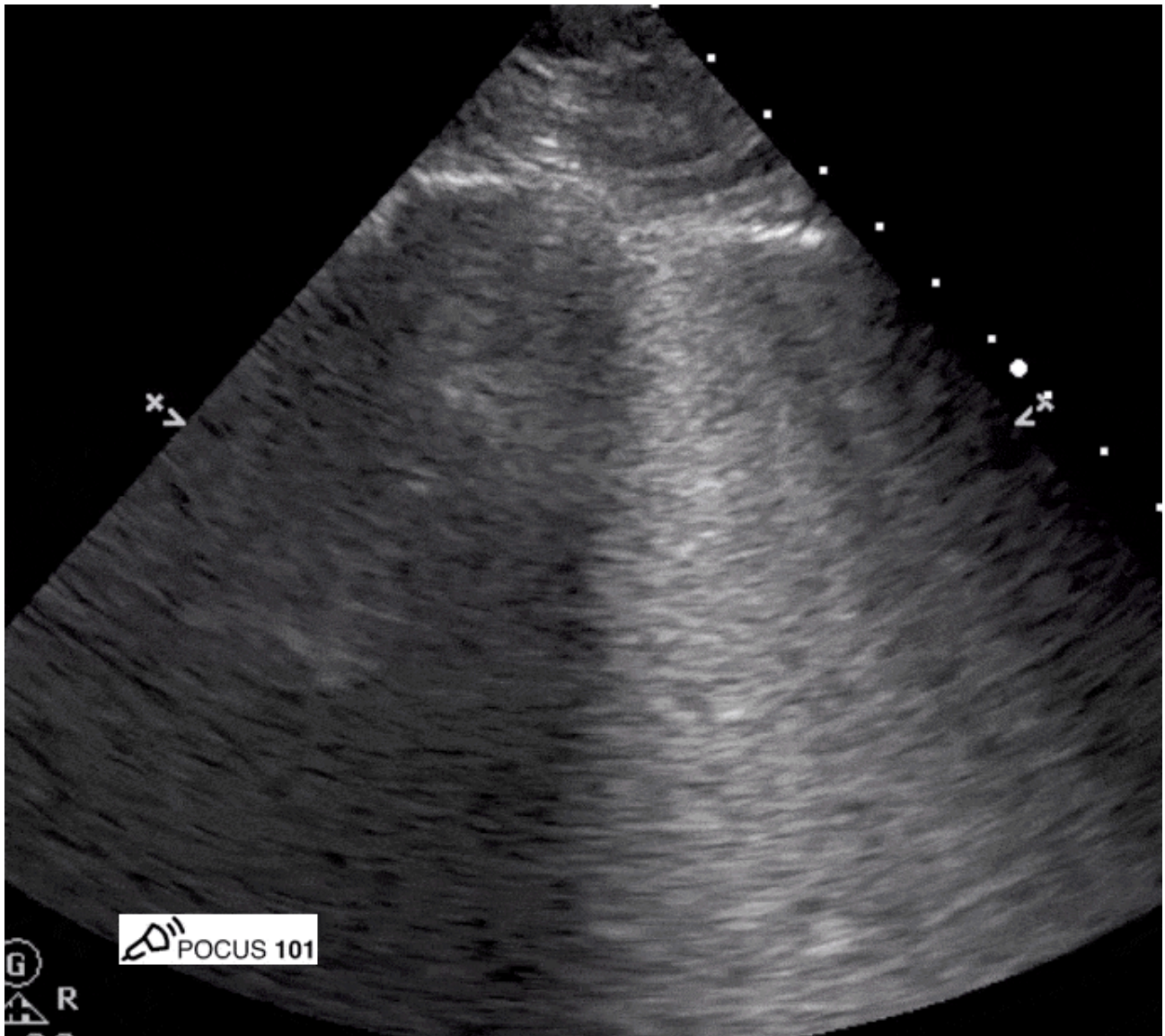


Few Scattered B-Lines

It is worth noting that patients with chronic scarring of their septa from pulmonary fibrosis, old infections, or interstitial lung disease have chronically widened/thickened septa that also create B-lines even *without* a “wet lung” etiology. Furthermore, lung fissures between the lobes can produce single B-lines, so less than 3 B-lines in a single field of view is generally considered normal.

Confluent B-lines

As more fluid builds up, it will become increasingly difficult to differentiate between singular B-lines. Thus, as more and more B-lines converge, they can create an appearance of “**Confluent B-lines**.”



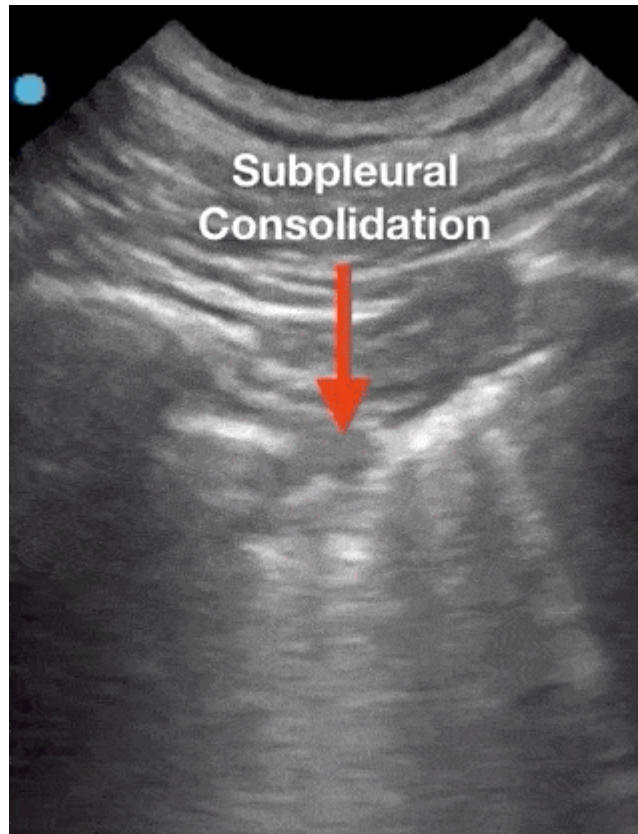
Confluent B-lines

Consolidations

As even more fluid builds up in the lung, parts of the lung can become *completely fluid-filled*, leading to **consolidation**. This is commonly seen in **pneumonia**. Note that consolidation can also be due to atelectasis from airway obstruction (i.e. mucous plug) or extrinsic compression (i.e. large pleural effusion).

As fluid build-up progresses, your ultrasound findings will progress from multiple B-lines, confluent B-lines, subpleural consolidation, the shred sign, to a dense consolidation. Once the air is completely gone from the lung and replaced with fluid this will result in an echogenic structure on ultrasound similar to echogenicity of the liver. This is termed **“hepatization of the lung.”**

Here is an example of **subpleural consolidation**:



Subpleural Consolidation

The Shred sign

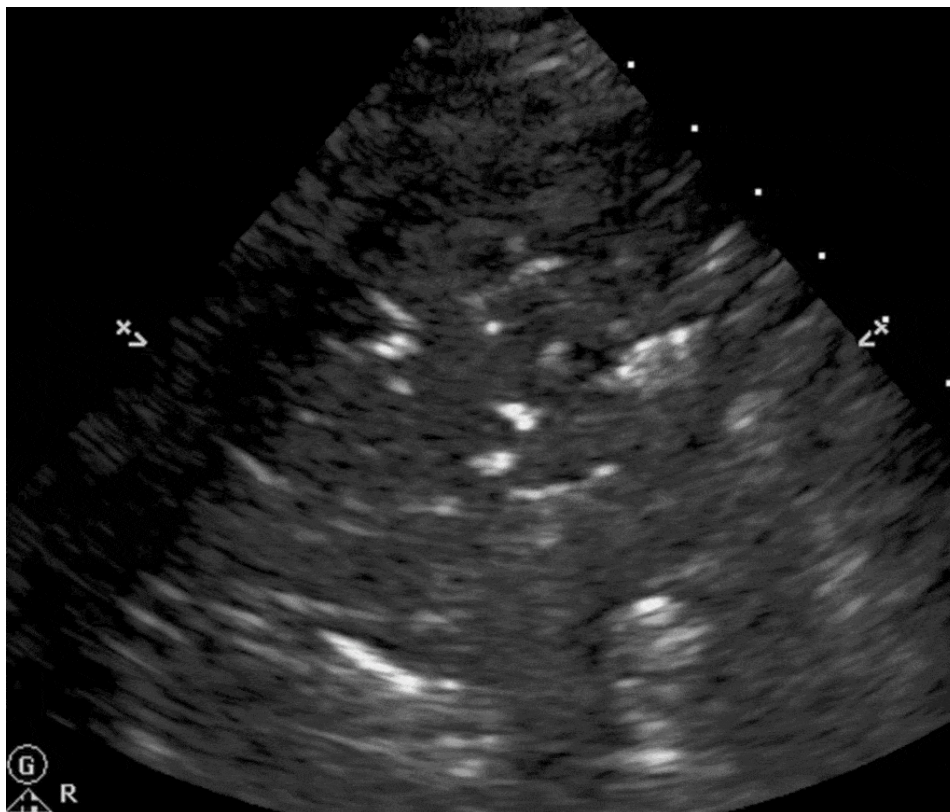
The border at which the fluid or pus-filled lung tissue meets air looks like a jagged echoic edge which is known as the “**Shred sign**” shown below (Lichtenstein 2009).



This image shows a subpleural consolidation abutting the pleural line (arrowheads) with a tissue-like pattern. This figure highlights the shredded lower border, limited by the arrows, which is the **shred sign**. (Lichtenstein 2009)

Dense Lung Consolidation

When lung is completely filled with fluid or is completely collapsed (atelectasis) you will have a dense appearing lung consolidation (Hepatization of the lung)

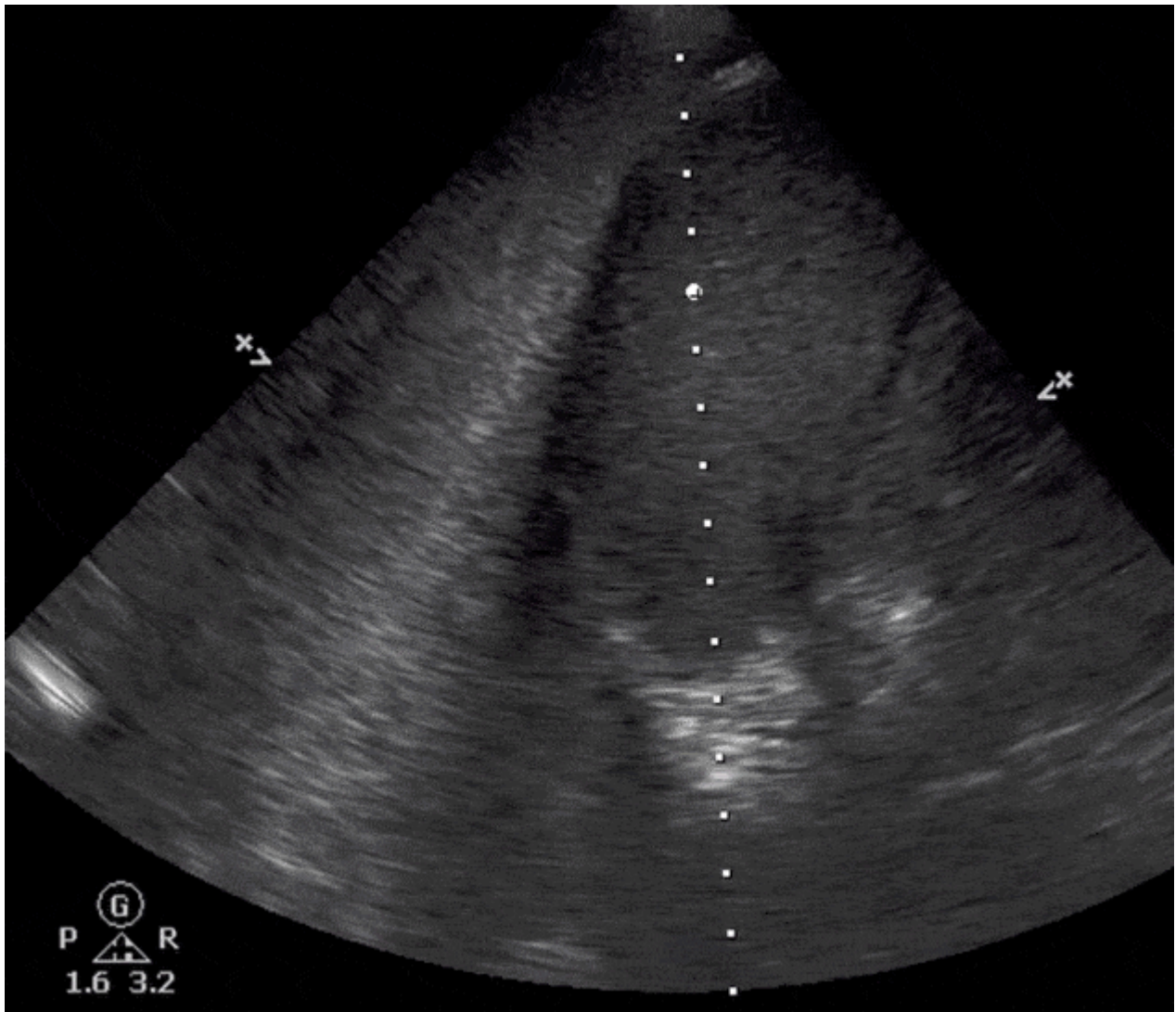


Dense Lung Consolidation

Air Bronchograms

Consolidations can also present with static and/or dynamic air bronchograms. On ultrasound, this trapped air appears as small hyperechoic specks.

- **Dynamic Air Bronchograms** tend to occur in pneumonia and move as the patient inhales and exhales.
- **Static Air Bronchograms** tend to occur when air bubbles are trapped behind an obstruction, as occurs in atelectasis, and don't move with respiration.

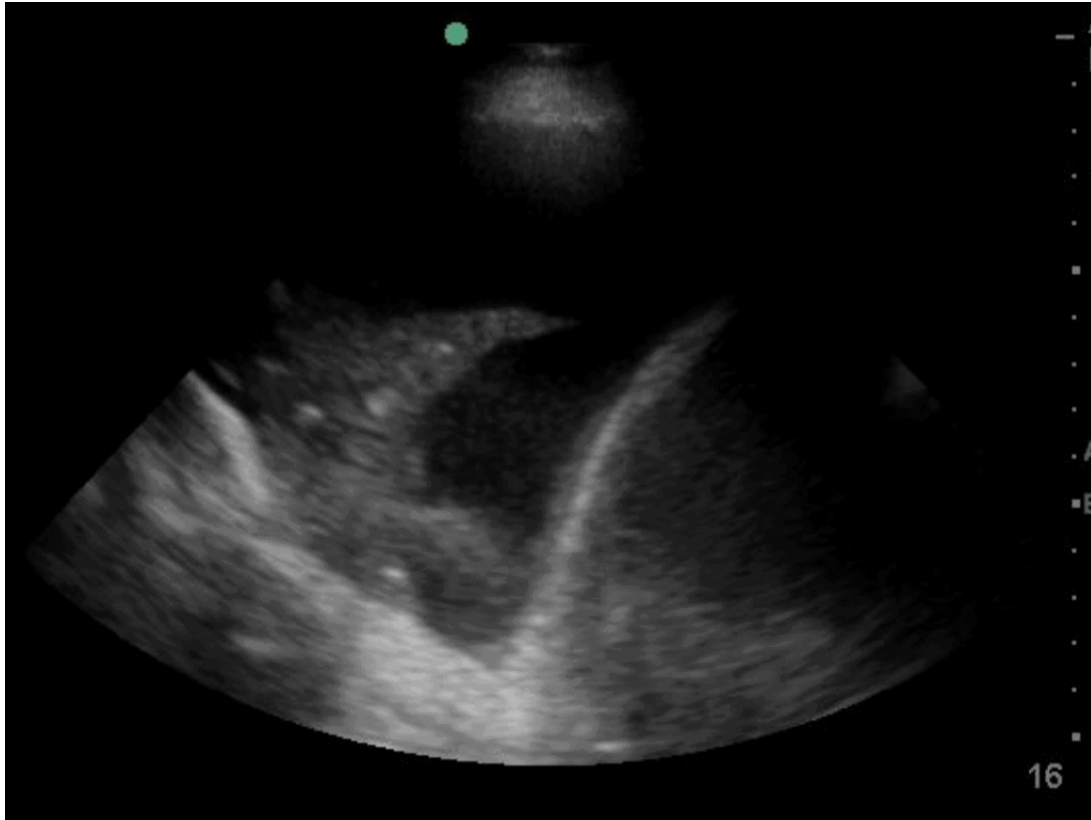


Dynamic air bronchograms at PLAPS point

Pleural Effusions

Pleural effusions occur when transudate or exudate builds up in the base of the lungs. They are best viewed from the PLAPS-point where the lower lobes are viewed. Point of Care Ultrasound (POCUS) is over 90% sensitive and specific for the detection of pleural effusions.

In this section you will learn how to recognize pleural effusions, calculate pleural effusion size, look for all of the common ultrasound signs.

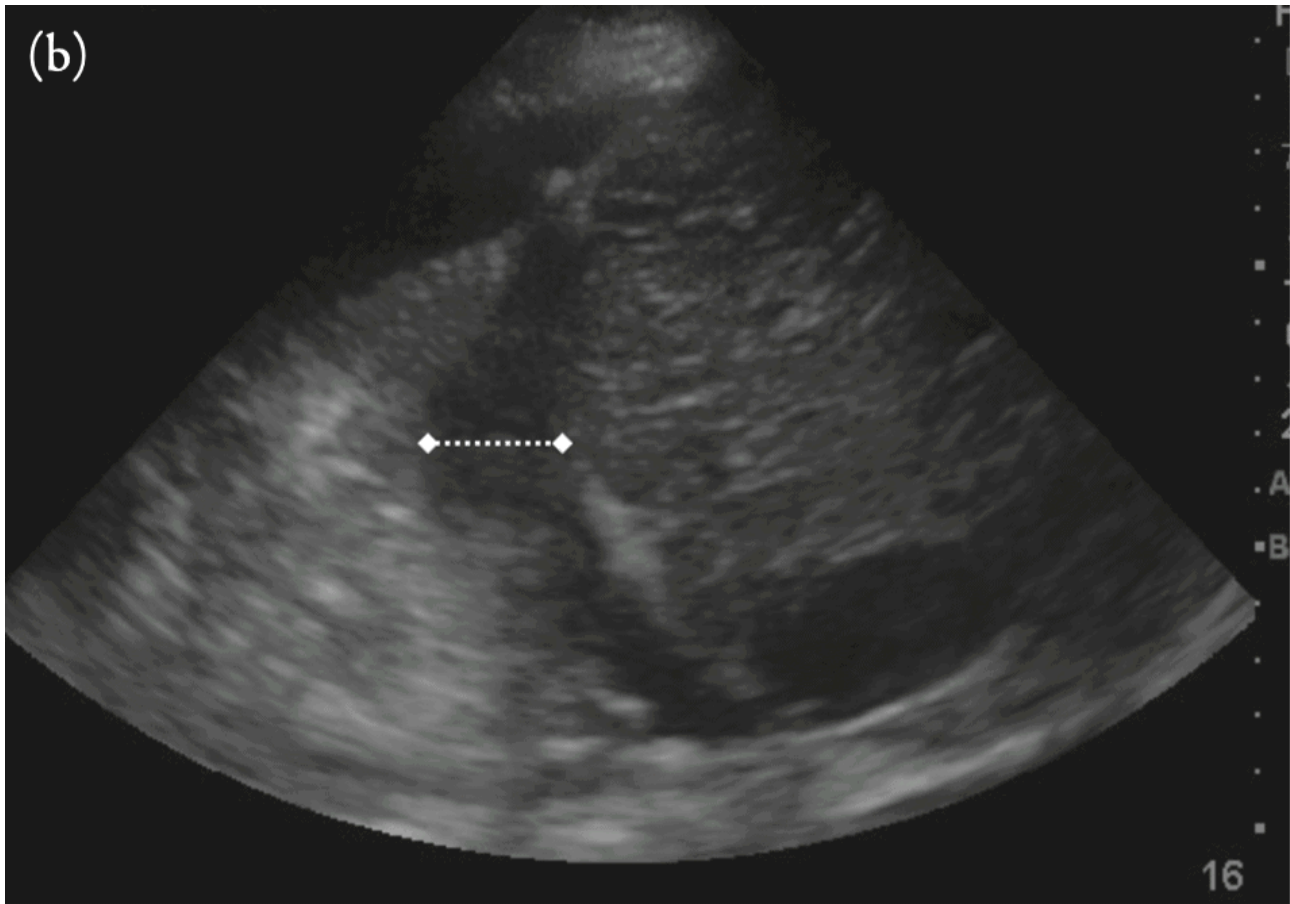


Pleural Effusion on Ultrasound

Calculating Pleural Effusion Size

The simplest way to estimate pleural effusion volume is to use the formula by Balik et al. It involves measuring the maximal effusion diameter (in millimeters) between the diaphragm and base of the lung in a supine patient (see Figure Below) and multiplying that number by 20.

$$\text{Pleural volume (mL)} = (\text{measured distance in mm}) \times 20.$$



Ultrasound Signs of Pleural Effusions

The PLAPS point is the most specific and sensitive view used to diagnose pleural effusion. If you have a patient with a suspected pleural effusion, the following signs/findings that can help you arrive at a proper diagnosis:

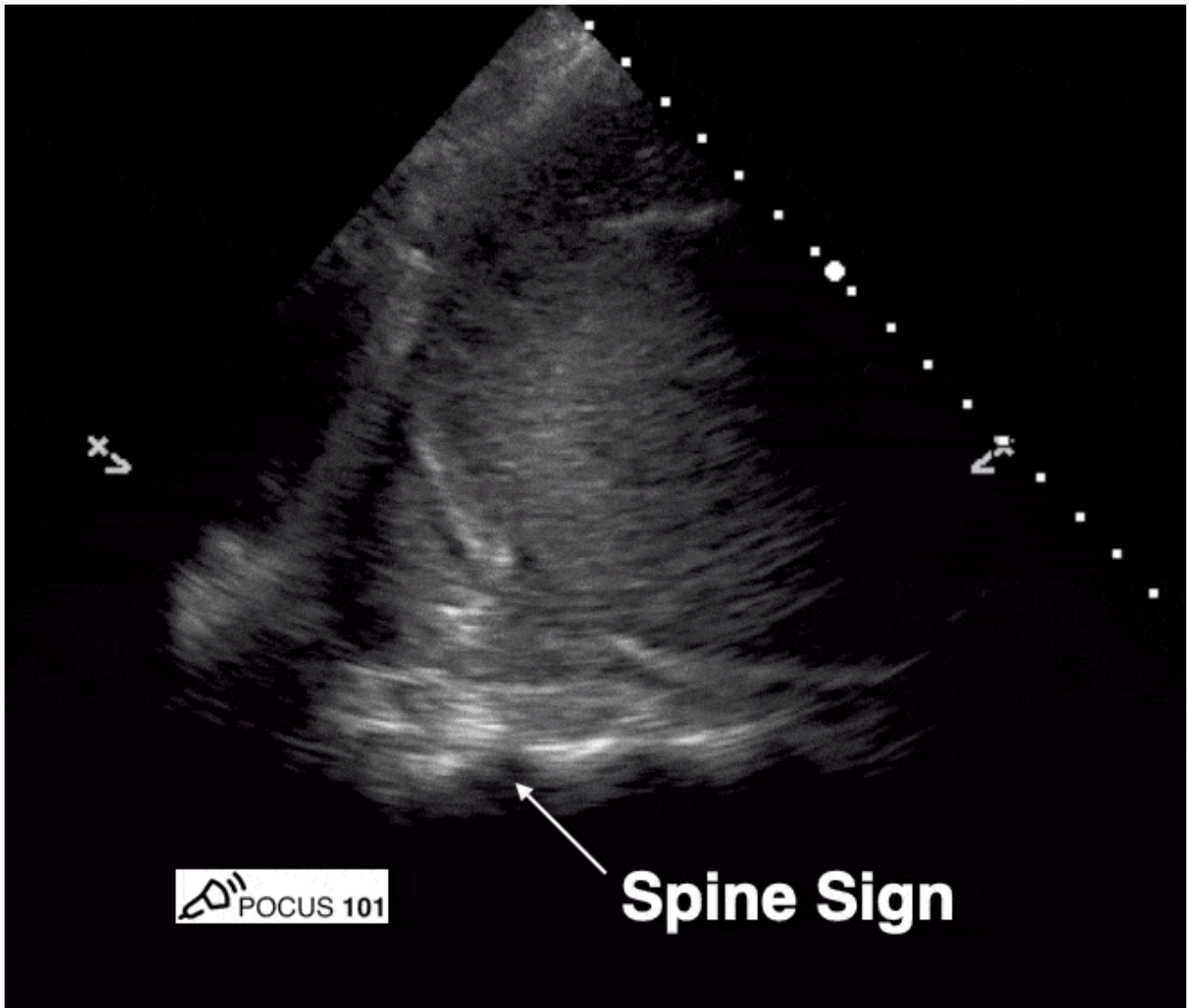
- Spine Sign
- Jellyfish sign
- Sinusoid Sign
- Quad Sign
- Plankton Sign
- Hematocrit Sign
- Loculated Pleural Effusions

Spine Sign

You will find a Spine sign at the PLAPS point along with a pleural effusion.

In a normal lung, you should be able to see the spine up until the edge of the diaphragm, but never passing the diaphragm. This is because the lung's air above the diaphragm prevents any sound waves from passing.

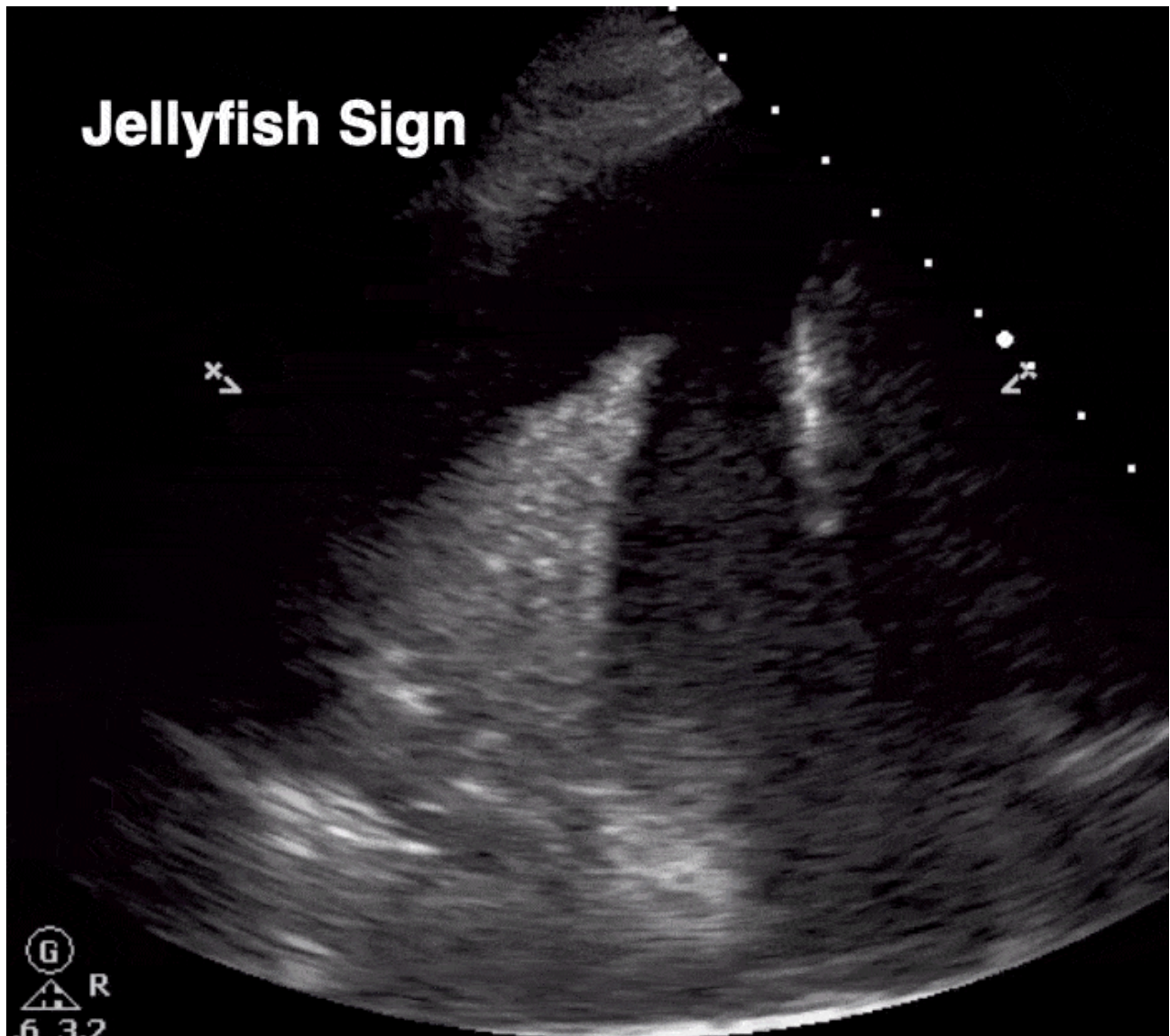
However, **in pleural effusions**, sound waves can pass through the pleural fluid allowing the spine to be seen above the diaphragm.



Spine Sign

Jellyfish Sign

You will find the Jellyfish sign at the PLAPS-point, located slightly above the diaphragm, in the RUQ. Recall that the PLAPS-point is the key location to find pleural effusions. The “**Jellyfish Sign**” occurs when a consolidated lung is seen floating in the pleural effusion.



Jellyfish Sign on Ultrasound

Sinusoid Sign

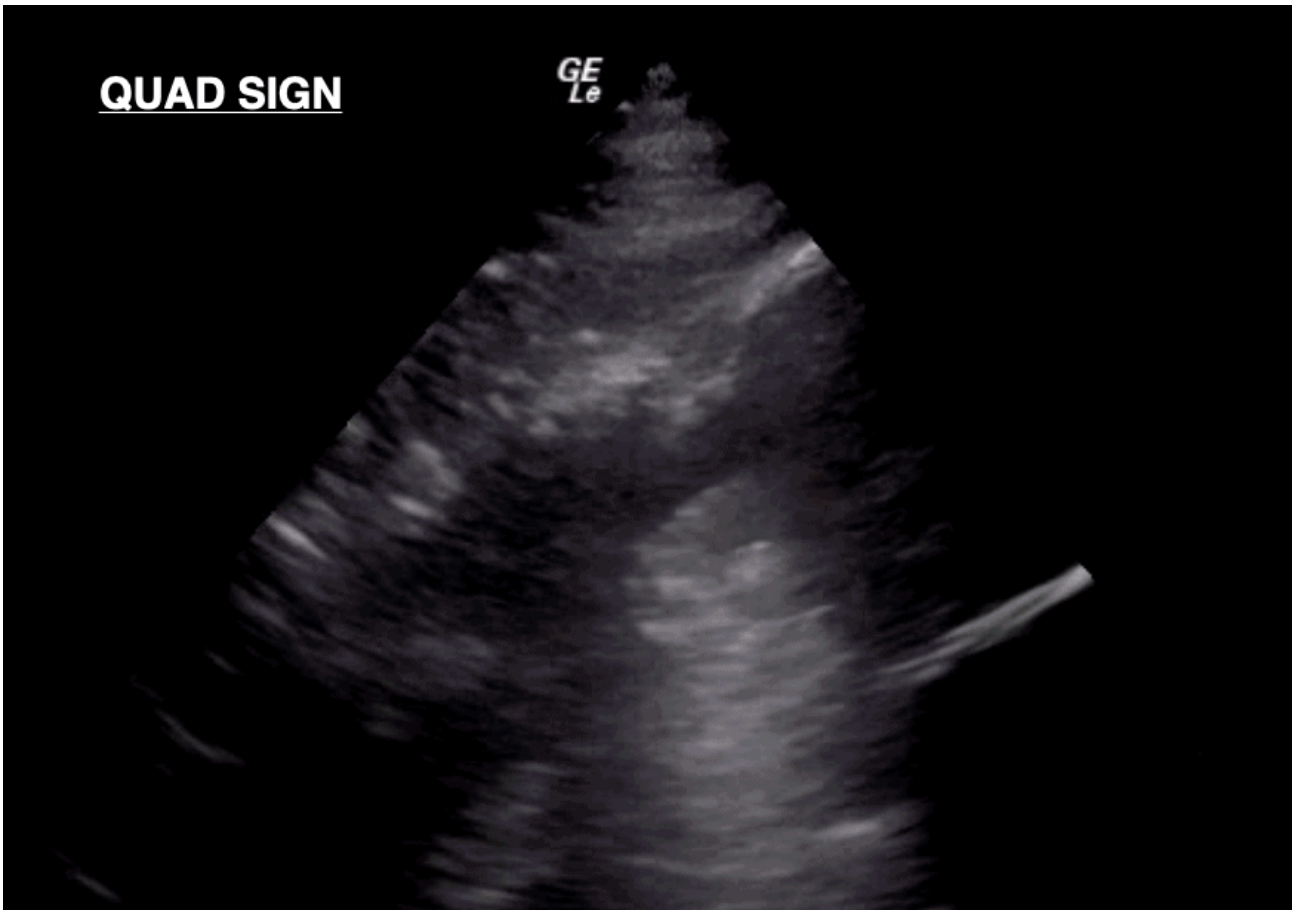
You can turn on M-mode to look for the **Sinusoid sign** (looks like a sine wave). It is caused by the parietal and visceral pleura moving closer and further apart while the patient breathes. (White arrows point to the lung line/visceral pleura while black arrows point to the pleural line/parietal pleura). It is equivalent to an M-mode view of the jellyfish sign.

Quad Sign

A pleural effusion has an anechoic appearance often delineated by the pleural line, the rib shadows, and the lung line, called the “Quad Sign.”

QUAD SIGN

GE
Le



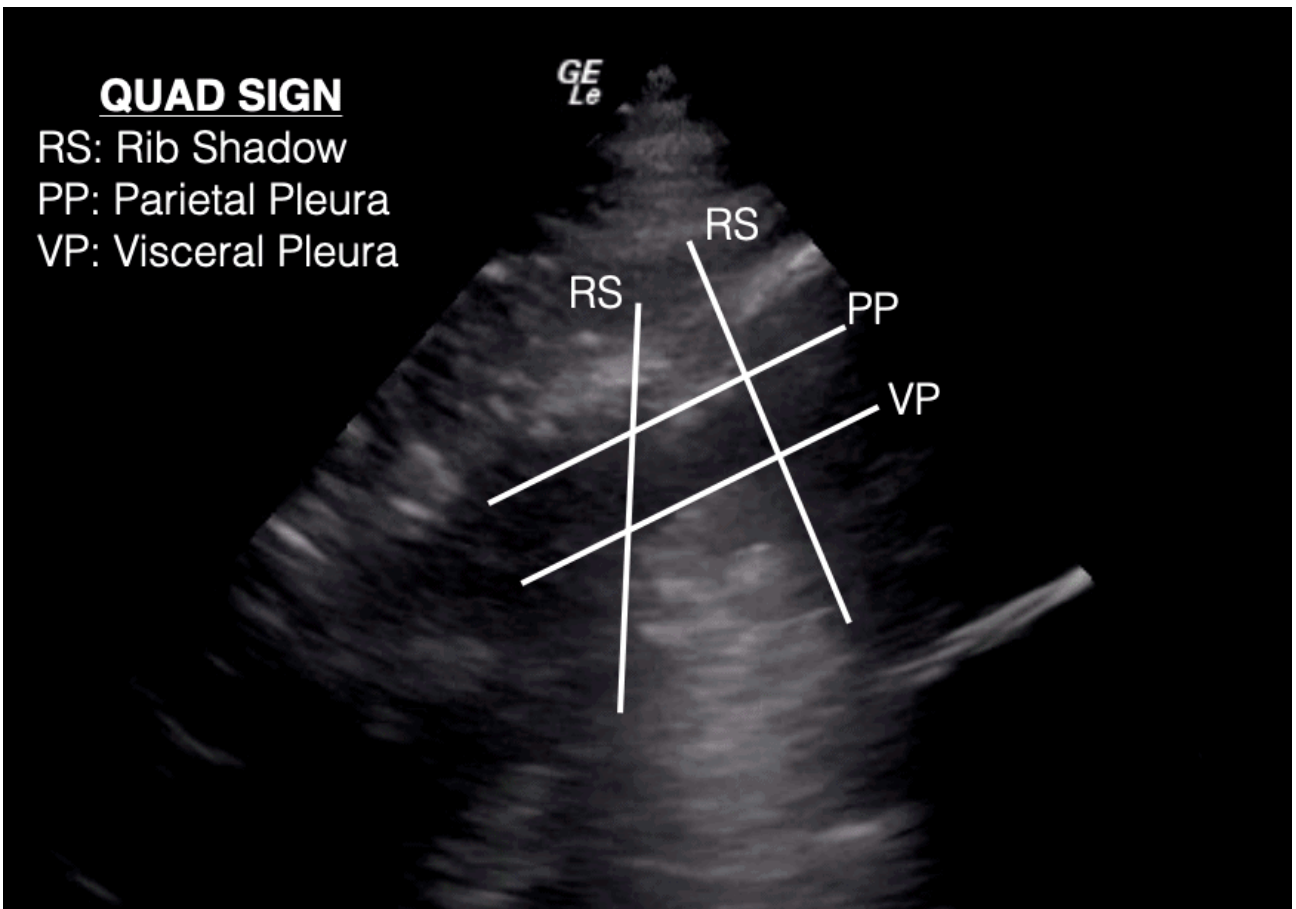
QUAD SIGN

RS: Rib Shadow

PP: Parietal Pleura

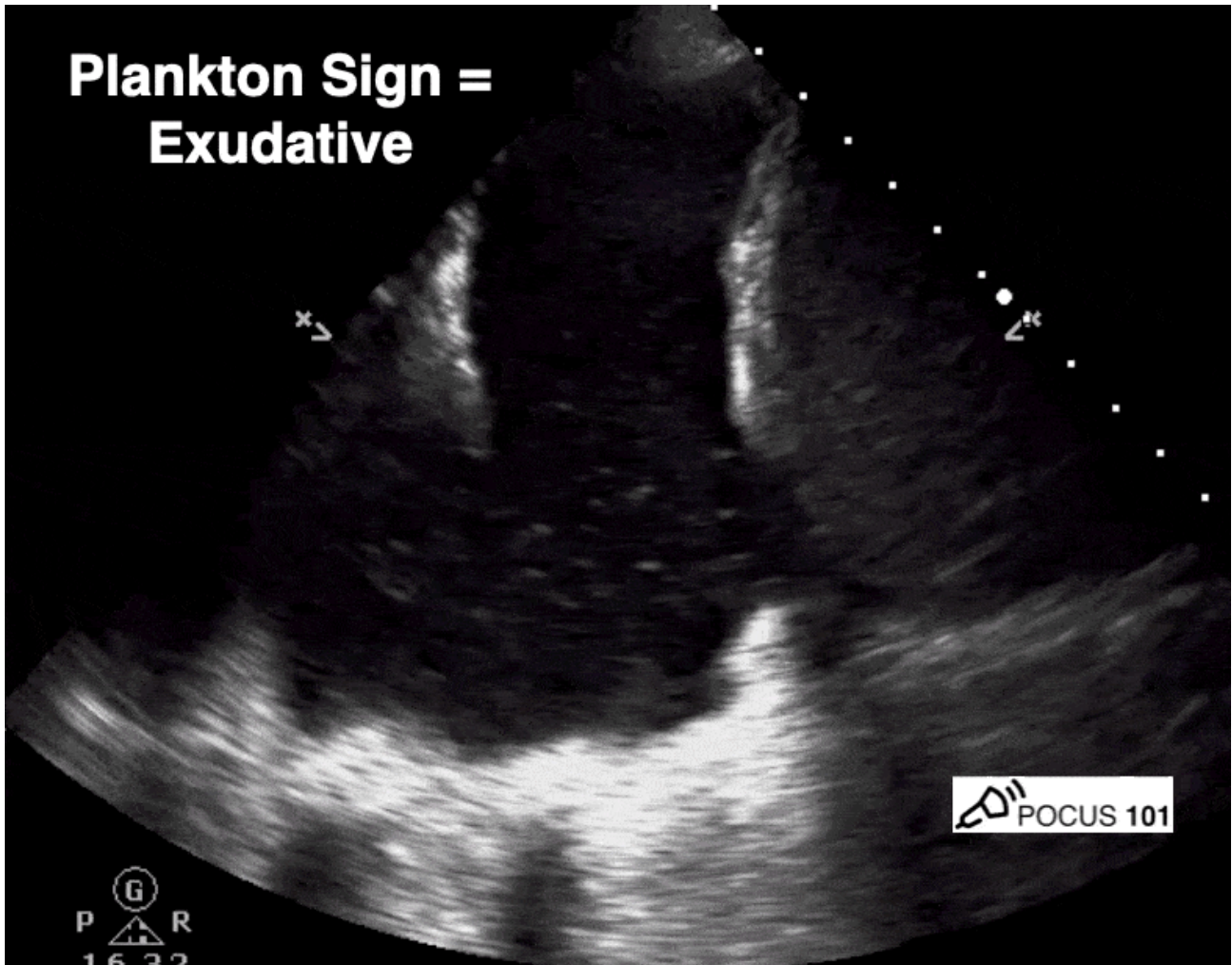
VP: Visceral Pleura

GE
Le



Plankton Sign

Now that you know how to detect a pleural effusion on point of care ultrasound (POCUS), you can further differentiate pleural effusions into transudative or exudative. Though most effusions are transudative, **exudative effusions can have the plankton sign**, as seen below. The plankton sign shows an effusion with **swirling, hyperechoic debris**.



Plankton Sign on Lung Ultrasound

Hematocrit Sign

The **hematocrit sign** refers to the echogenic layering of material in a pleural effusion. This can be due to exudative effusions or a hemothorax (Chichra A).



Hematocrit Sign on Lung Ultrasound

Transudative vs Exudative Pleural Effusion

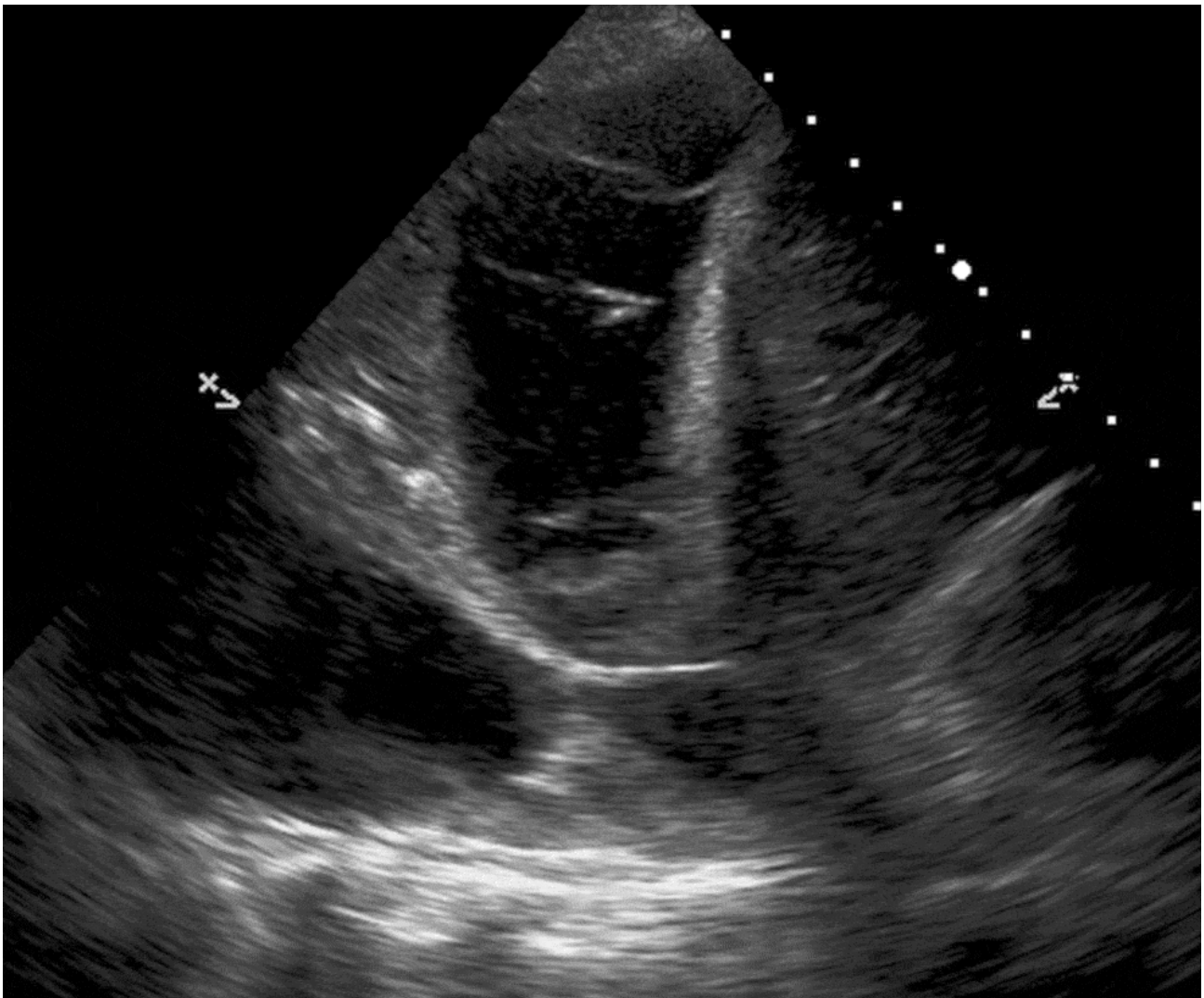
Here are some differences between transudative and exudative pleural effusions:

	Transudative Pleural Effusion	Exudative Pleural Effusion
Ultrasound Findings	– Absent plankton sign/hematocrit sign– Typical effusion signs	– Present plankton sign/hematocrit sign– Typical effusion signs
Fluid Quality	– Transudate with low protein and cell count – Normal glucose levels	– Cloudy transudate with high protein and cell count – Light's Criteria: Pleural <i>fluid protein</i> >0.5 and LDH >0.6 or LDH– Low glucose levels
Mechanism	– Increased capillary hydrostatic pressure and/or decreased capillary oncotic pressure (low albumin).	– Increased vascular permeability from inflammation/infection/malignancy
Common Etiologies	– Heart failure, nephrotic syndrome, liver cirrhosis	– Trauma (hemothorax), bacterial infections, lymphatic obstructions, malignancies, pulmonary embolism

Loculated Pleural Effusion

Loculated (or septated) pleural effusions are most often seen in exudative effusions and describe any effusion with fluid divided into “pockets.” They can be caused by infections, abscesses, scarring, or fibrosis in the pleural cavity that complicates proper fluid drainage.

The image below shows an image of a loculated pleural effusion that lies above the diaphragm.



Loculated Pleural Effusion

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Major Lung Ultrasound Pathology Profiles

Now that you have a grasp of the most common pathological lung signs to look for, this section will help you use those findings to characterize more specifically the ultrasound profiles of **pneumothorax, pneumonia, cardiogenic pulmonary edema, acute respiratory distress syndrome, and COPD**.

You can choose to use an algorithm such as the Blue Protocol (Lichtenstein, et al) but after teaching many learners lung ultrasound we've found it much easier for you to recognize the lung ultrasound findings that go along with each disease pattern. We will present all of the major disease findings and give you examples of the corresponding lung ultrasound findings below.

Lung signs/findings specific to each disease will be further explained within their respective section. Keep in mind, these are the *potential* Lung ultrasound findings you will see at the

bedside, however, you must coordinate these profiles clinically with what is going on with the patient.

Pulmonary Pathology	Lung Ultrasound Findings
Pneumothorax	– No lung sliding/barcode sign– Lung point (hard to find)– Bilateral A-lines (parietal pleura reflection)
Pneumonia	– Consolidation or “hepatization of the lung”– Supleural Consolidations– “Shred” sign– Unilateral B-linesor bilateral B-lines– Possible small plural effusion– Reduced lung sliding due to thickened pleura in severe cases– Dynamic air bronchograms (atelectasis has static bronchograms)
Cardiogenic Pulmonary Edema	– Widespread and bilateral B-lines– Normal lung sliding– Possible bilateral effusion depending on the severity
Acute Respiratory Distress Syndrome (ARDS)	– Bilateral or unilateral B-lines– Normal lung sliding
COPD or Asthma	– Bilateral A-lines– Reduced lung sliding
Pulmonary Embolism (PE)	– Bilateral A-Lines– Deep Vein Thrombosis in Upper or Lower Extremities– Right Ventricular Enlargement (massive/submassive PE)

Pneumothorax

Ultrasound is incredibly useful for diagnosing pneumothorax with high sensitivity and specificity. A pneumothorax occurs when a lung collapses due to loss of negative pressure between the visceral and parietal pleurae. This **abolishes lung sliding**.

However, there is usually a location where sliding still exists. The transition from sliding to no sliding is known as the **lung point sign**. Though difficult to find, a lung point is 100% specific for ruling in a pneumothorax (Chan S.).

Clinical manifestations of pneumothorax include pleuritic chest pain, dyspnea, tachycardia, and reduced breath sounds on the affected side.

Ultrasound Profile of Pneumothorax

- No lung sliding
- M-mode barcode sign

- Lung point Sign
- A-lines from intact parietal pleura

Here are three important steps to evaluating for pneumothorax:

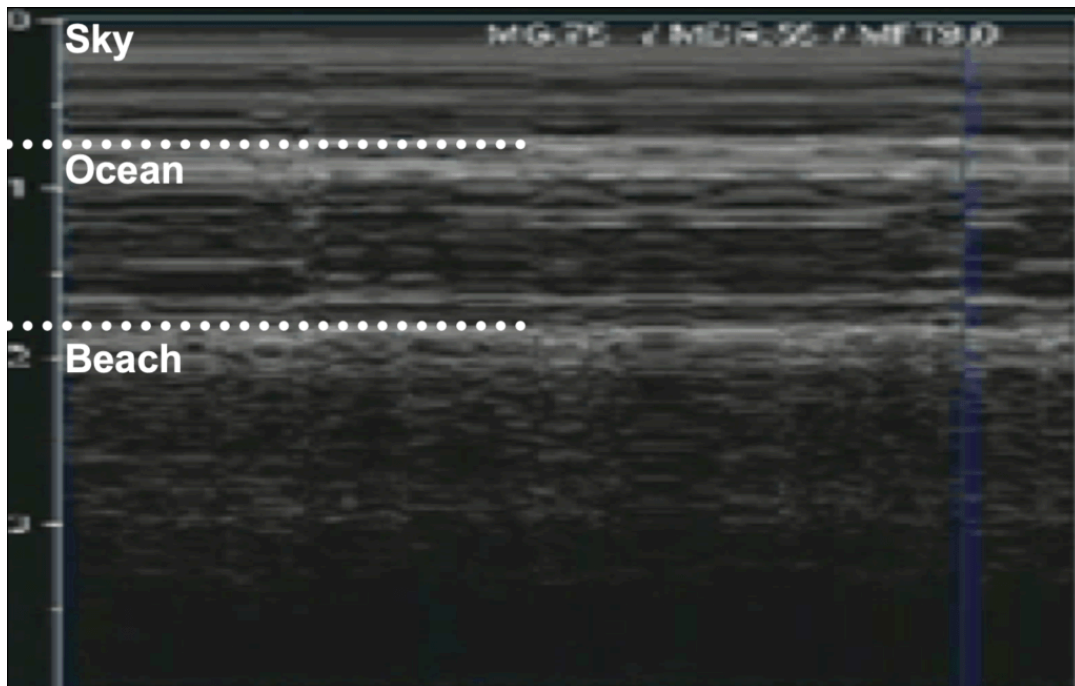
First, if lung sliding is present, you can rule *out* pneumothorax with 100% accuracy at that ultrasound point (Husain LF).

Remember that presence of lung sliding only rules out pneumothorax at that specific point you are scanning. Make sure to maximize your sensitivity by scanning multiple points on the chest.

You can look for lung sliding with B-mode or M-mode:



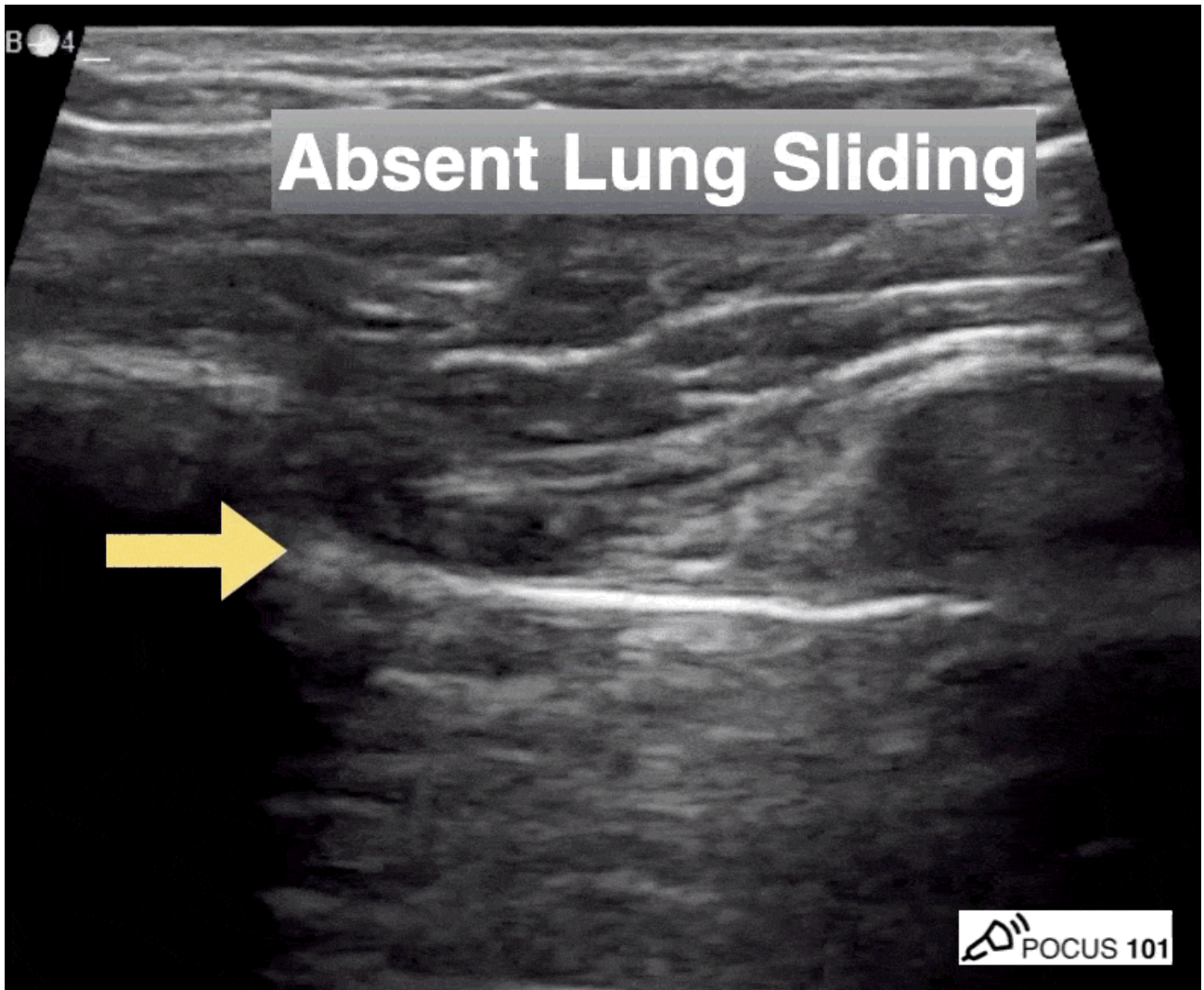
Normal Lung Sliding (B-mode)



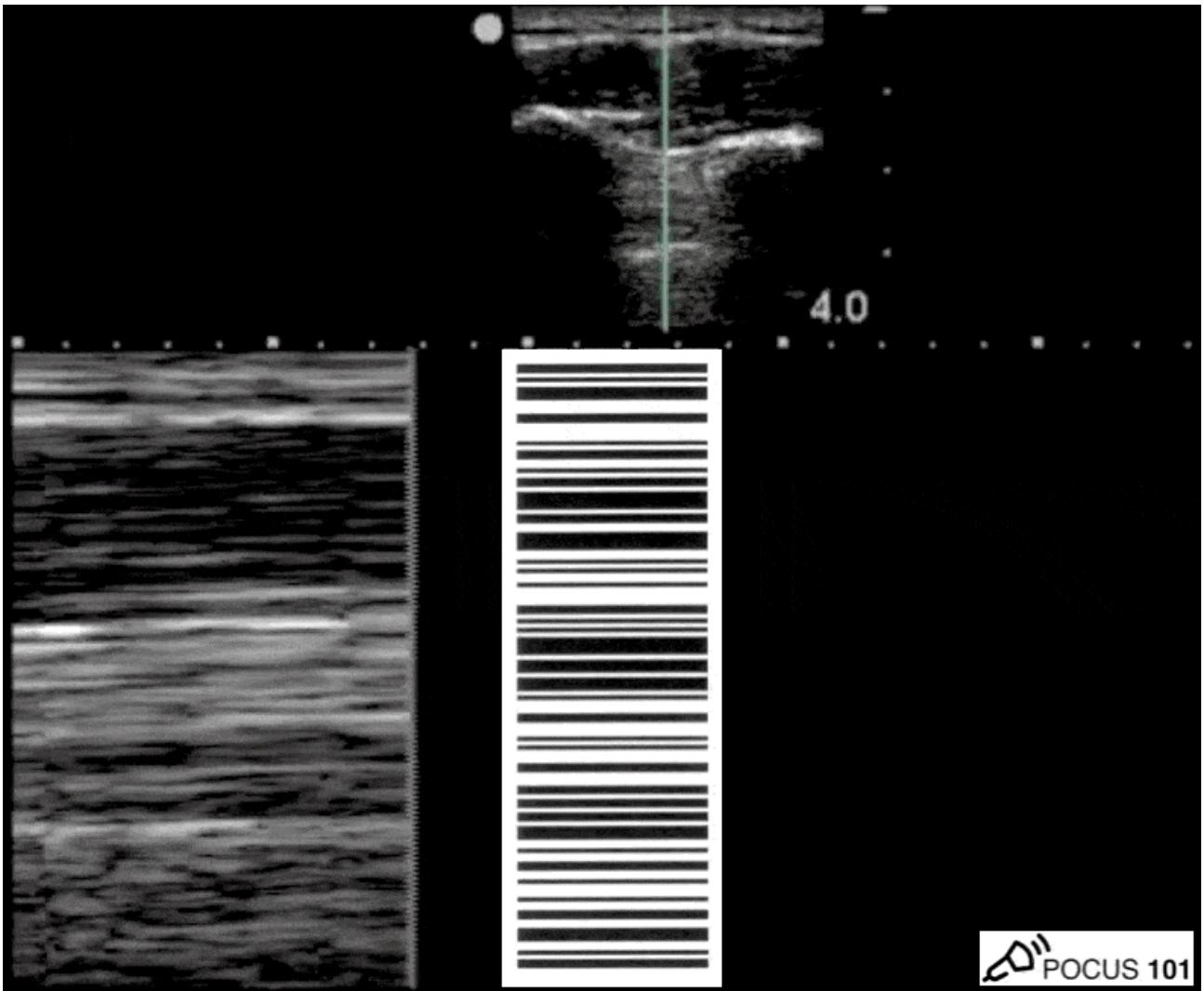
Normal Lung Sliding with Seashore sign (M-mode)

Second, if lung sliding is **ABSENT, you should *not automatically assume* pneumothorax.**

Recall other causes of reduced/absent lung sliding: severe consolidation, chemical pleurodesis, acute infectious or inflammatory states, fibrotic lung diseases, acute respiratory distress syndrome, or mainstem intubation.



Absence of Lung Sliding (B-mode)



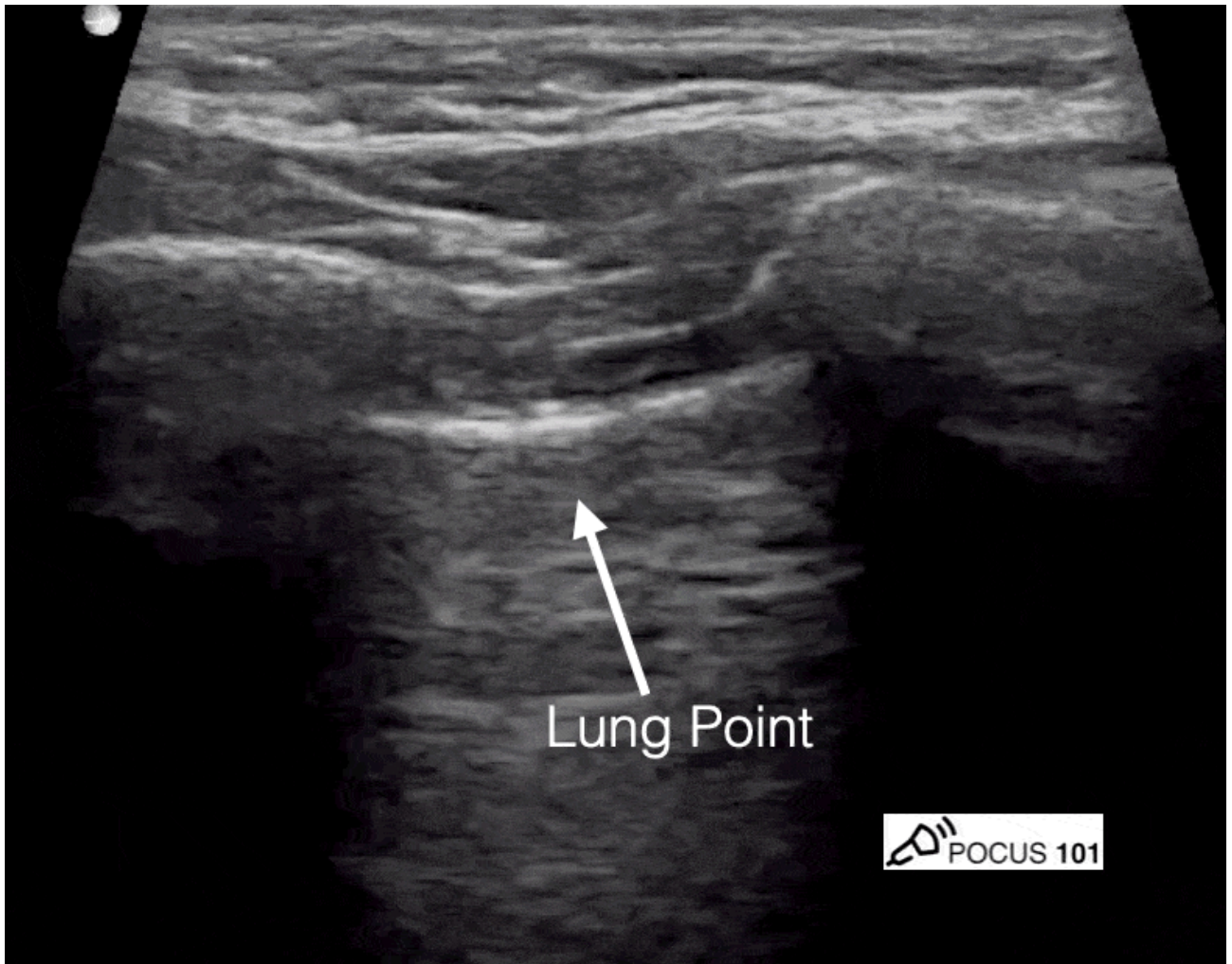
Absence of Lung Sliding – Barcode Sign (M-Mode)

Third, if a lung point is present, you can rule *in* pneumothorax with 100% accuracy (Chan S).

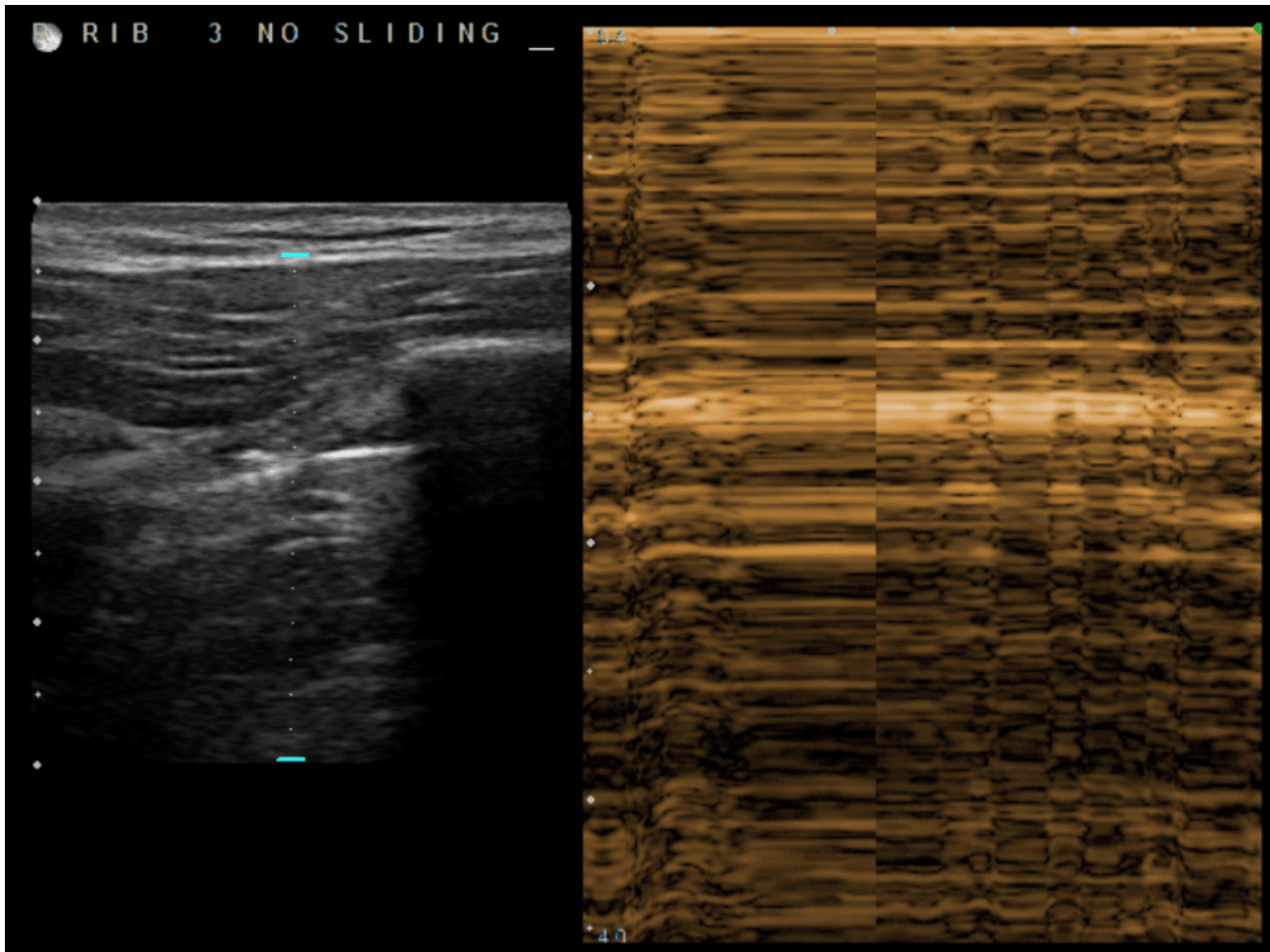
To confirm the presence of a pneumothorax, you should look for the “**Lung Point Sign.**”

The **lung point** is when you can see the transition between normal lung sliding and the absence of lung sliding. This is the transition point between the collapsed lung and normal lung. If you see this you can definitively *rule in* a pneumothorax. The Lung point sign also helps you quantify how large a pneumothorax is.

If you think you may have found a lung point but are not sure, use M-Mode and place your cursor at the intersection where you think lung sliding starts and stops. If you see a normal seashore sign that turns into an abnormal barcode sign, then you have located the lung point with M-Mode.



Lung Point Sign (B-mode)



Lung Point (M-mode)

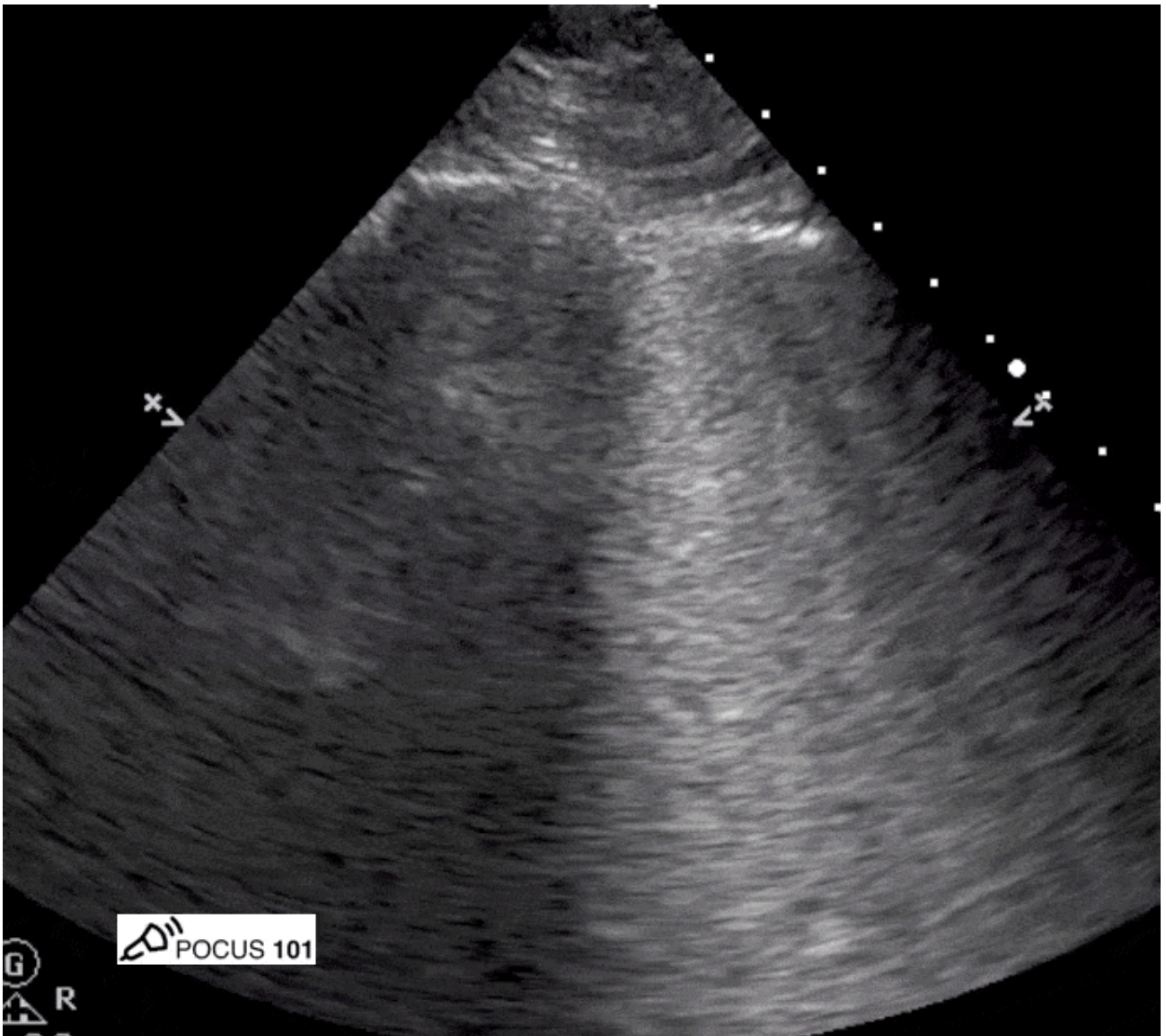
Pneumonia

Pneumonia is an infection that inflames the alveoli and fills them with fluid. This extra fluid creates **unilateral or bilateral B-lines depending if one or both lungs are affected.**

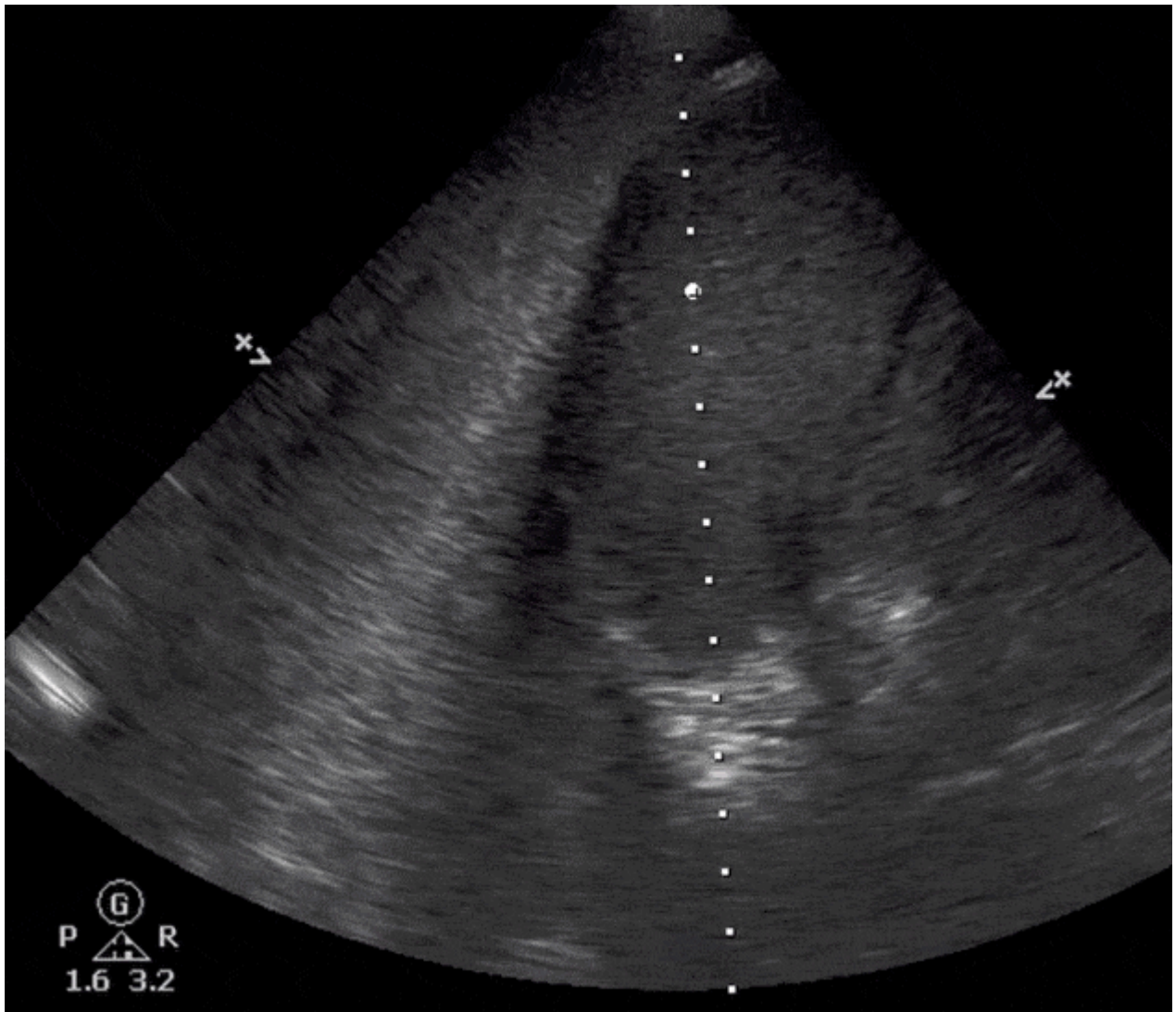
The fluid or pus-filled alveoli also cause a productive cough with colored phlegm, fever, chills, and difficulty breathing.

Ultrasound Profile of Pneumonia:

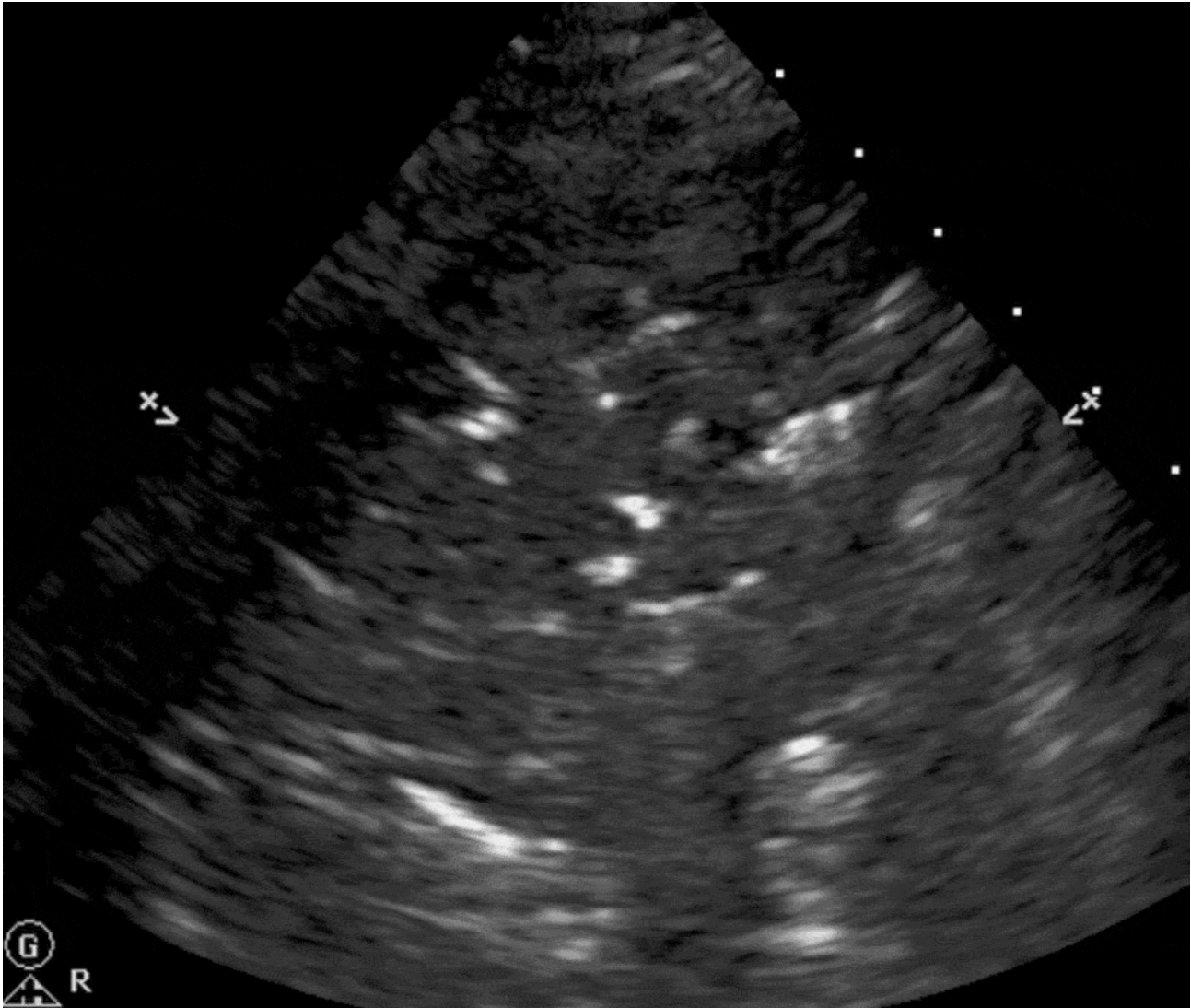
- Shred sign
- Consolidated lung tissue with dynamic air bronchograms
- B-lines: unilateral (bacterial) or bilateral (viral)
- Decreased lung sliding depending on severity
- Small pleural effusion depending on severity



Confluent B-lines



Consolidation with Dynamic Air Bronchogram



Consolidated Lung

Even the most experienced ultrasound practitioners have difficulty distinguishing pneumonia from atelectasis using ultrasound alone. Thus, a clear clinical picture is necessary before determining the true cause of lung consolidation. The table below outlines some of the key differentiating factors between pneumonia and atelectasis.

	Pneumonia	Atelectasis
US Findings	– Consolidation pattern – Shred sign– - Dynamic bronchograms– Possible pleural effusion (parapneumonic effusion)	– Consolidation pattern – - Static air bronchograms
Distinguishing clinical features	– Fever and chills– Cough with colored mucous or blood– Severe Malaise– Crackles and breath sounds	– Obstruction to airway– Acute dyspnea– Mucous plug in Bronchus

Cardiogenic Pulmonary Edema (CPE)

Ultrasound is useful for assessing the severity of cardiogenic pulmonary edema.

Cardiogenic pulmonary edema is a type of **pulmonary edema** caused by increased pressures of the left side of the heart. This condition often occurs secondary to congestive heart failure when the left ventricle cannot pump out all the blood it receives from the lungs. **The backed-up blood increases capillary hydrostatic pressure that then causes fluid to leak into both lungs, resulting in bilateral and symmetric B-lines.**

As CPE progresses, more B-lines appear until they converge into vertical sheets, known as **confluent B-lines**. As fluid builds up further, expect to see bilateral transudative pleural effusions with atelectasis of the lungs. Cardiogenic pulmonary edema usually affects both lungs but can atypically affect a unilateral lung.

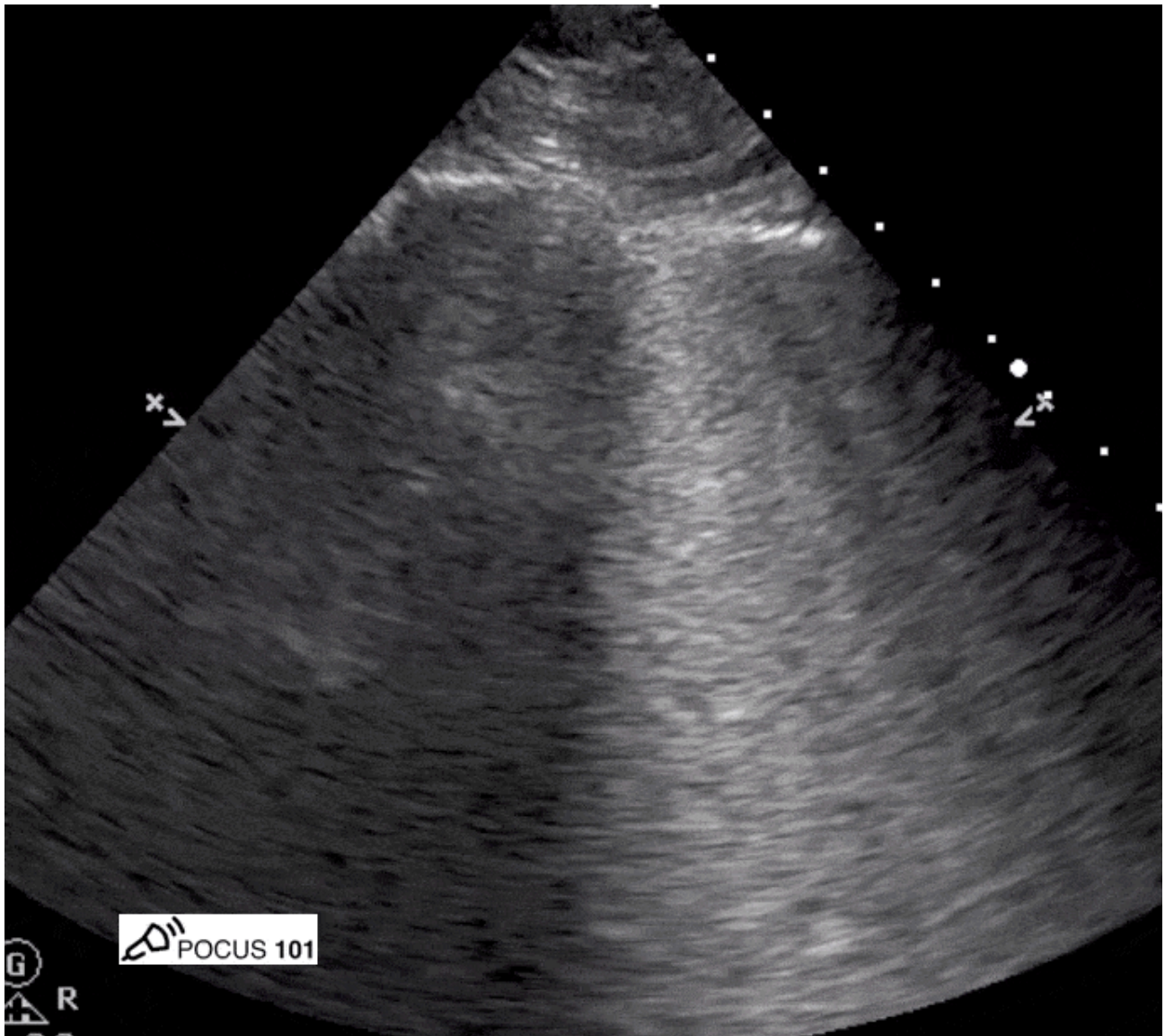
If you suspect your patient has CPE, it is also recommended to perform an ultrasound evaluation of their heart to assess for systolic and [diastolic dysfunction](#) of the left ventricle.

Cardiogenic Pulmonary Edema – Lung Ultrasound Profile:

- Bilateral and Symmetric B-lines (can sometimes be unilateral)
- Confluent B-lines with increasing severity
- Pleural edema and/or bilateral effusions with increasing severity
- Lung Sliding Present



Severely Decreased Ejection Fraction



Confluent B-Lines

Acute Respiratory Distress Syndrome

Acute respiratory distress syndrome (ARDS) is often fatal and the risk of death increases with age and severity of the illness. ARDS is characterized by the rapid onset of widespread inflammation in the lungs. **The most common cause of ARDS is sepsis**, but can also be precipitated by inhalation of harmful substances, pancreatitis, aspiration, severe pneumonia, or near-drowning experiences. Notice how nearly all of these etiologies involve a form of fluid buildup or inflammation so ARDS will present with **B-lines** on thoracic ultrasound.

People with ARDS also have severe shortness of breath, rapid breathing, and may often need support from a ventilator.

Since the following ultrasound profile is similar to the profile of cardiogenic pulmonary edema, practitioners must use other clinical features to interpret these findings to rule out elevated left heart pressure as the cause of the B-lines.

ARDS Ultrasound Profile

- Bilateral or unilateral B-lines
- Lung sliding present (but may be diminished due to inflammation)
- On echocardiography, the patient should have normal [diastolic function](#).

Chronic Obstructive Pulmonary Disease/Asthma

Chronic obstructive pulmonary disease (COPD) is a collection of lung diseases that obstruct airflow. This group of diseases can include asthma, emphysema, and chronic bronchitis. Patients with COPD or severe asthma often have hyperinflated lungs. If hyperinflation is severe enough, the resultant shallow breathing can reduce lung sliding.

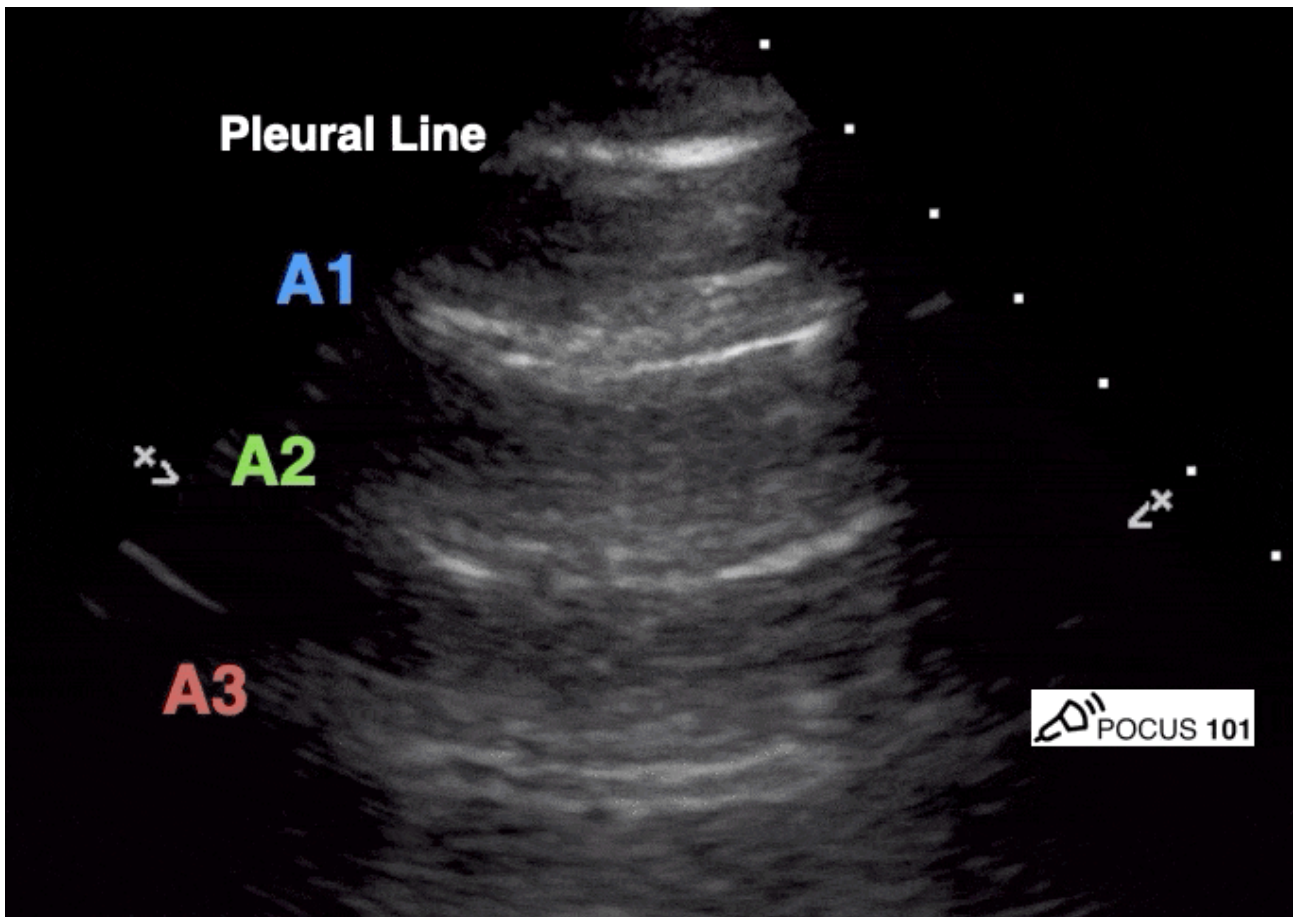
If you see normal findings such as **lung sliding and A-lines** but the patient still has symptoms and difficulty breathing, you should consider COPD, asthma, pulmonary embolism, or nonpulmonary conditions causing the patient's dyspnea.

COPD/Asthma Ultrasound Profile

- Bilateral A-lines
- Normal sliding or reduced lung sliding (as seen in severe asthma/COPD)



Lung Sliding

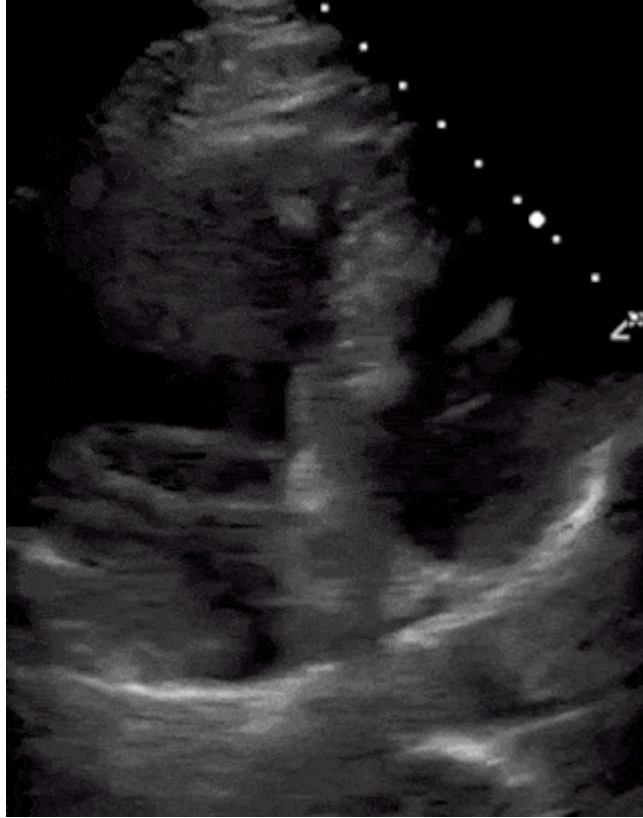


A-lines caused by pleural membrane reflections.

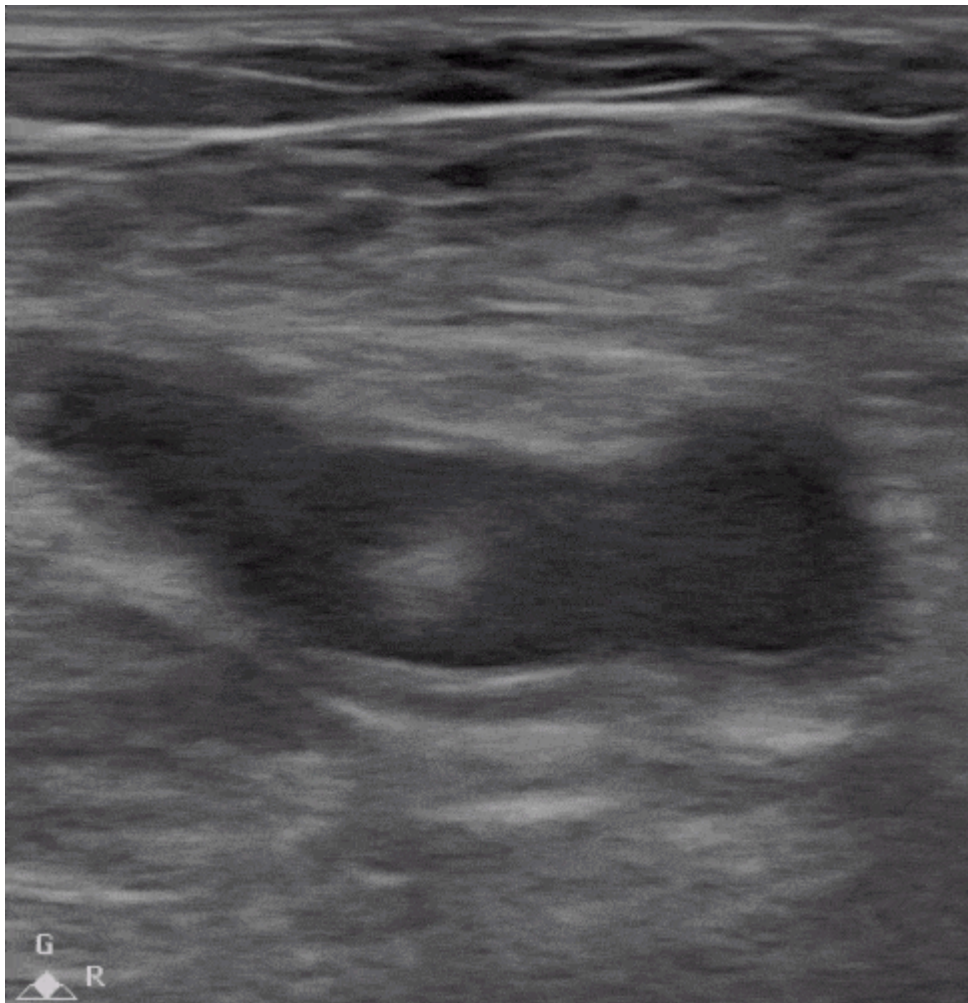
Pulmonary Embolism

Pulmonary embolism is often a difficult and missed diagnosis. Unfortunately lung ultrasound will usually appear normal (lung sliding with A-lines) in patients with pulmonary embolism.

If you are concerned for pulmonary embolism, perform a cardiac ultrasound to look for [right ventricular strain](#) and a lower extremity [DVT \(deep vein thrombosis\) scan](#) rule out venous thromboembolism. If those ultrasounds are normal consider obtaining a CT angiogram of the chest to look for pulmonary embolism.



RV Strain with McConnell's Sign



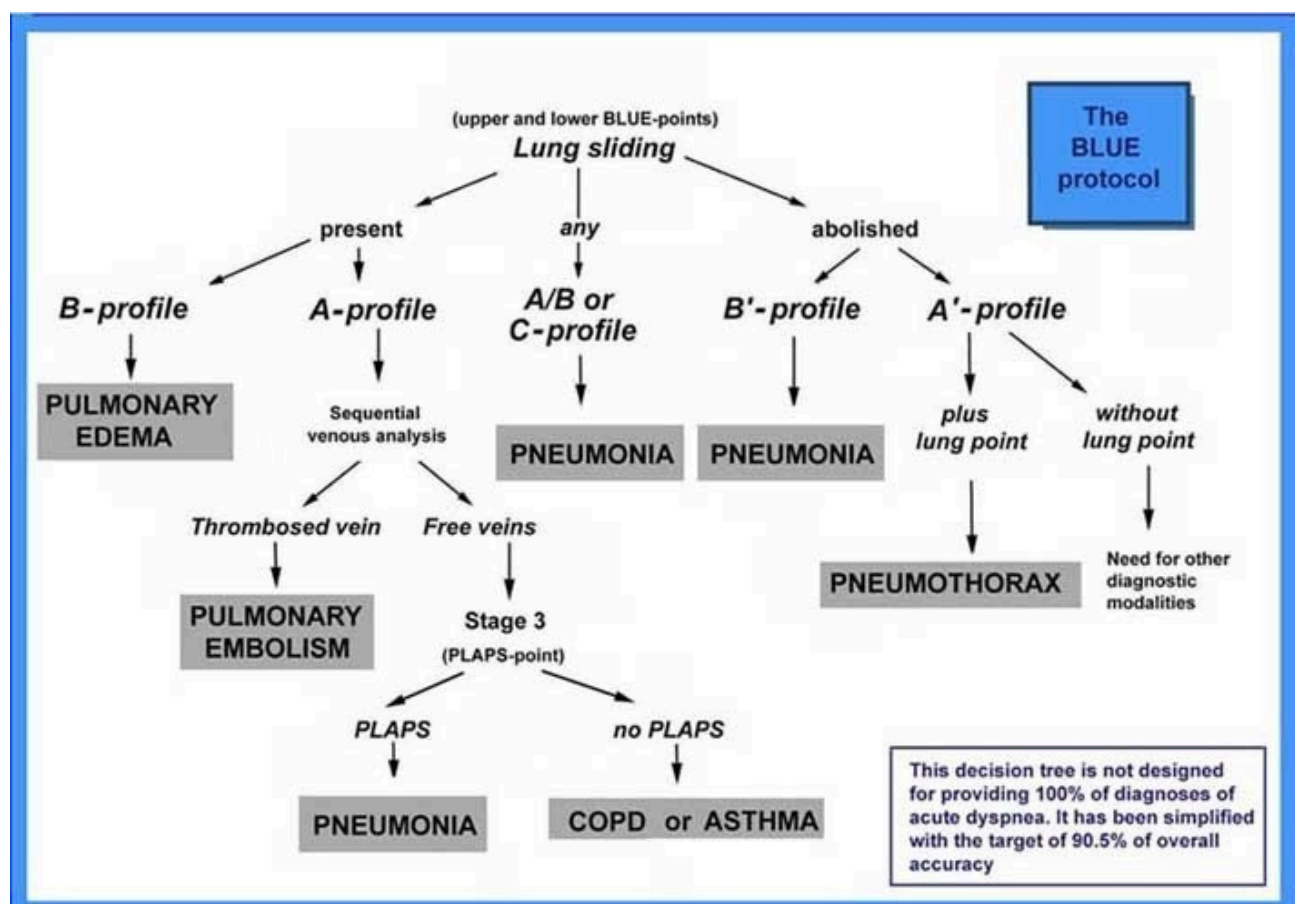
Mobile DVT in Lower Extremity

The Blue Protocol

Researchers have created protocols to streamline the use of pulmonary ultrasound in clinical practice. The most widely used protocol has been the BLUE protocol created by Lichtenstein and colleagues and is used by many practitioners to diagnose the cause of acute respiratory failure. Pulmonary edema, pulmonary embolism, pneumonia, chronic obstructive pulmonary disease, asthma, and pneumothorax yield specific profiles predicted by the BLUE protocol.

This protocol classifies A-lines and B-lines into various profiles. Though quite simple, they can be confusing if you are not first familiar with them.

- “A-profile” has A-lines with lung sliding and describes normal lungs
- “B-profile” has B-lines and lung sliding
- “B’ profile” has B-lines with **impaired** lung sliding
- “A’ profile” has A-lines with **impaired** lung sliding
- A/B profile has one lung with A-lines and the other with B-lines.



(Lichtenstein, 2008)

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